Date 17-07-2025

Expt. No. _____ Page No. _____

UNIT-1

Syllabus :-

Introduction : Network Hardware, Network Topologies, Reference models - The OSI Reference model - the TCP/IP reference model - A Comparison of the OSI and TCP/IP Reference, Basics of Packet Switching, Circuit Switching & Virtual circuit switching, models.

Physical layer : Guided transmission media, Twisted pairs, Coaxial cable, Fibre optics, wireless transmission.

A computer network is a collection of interconnected devices which share resources & information. These devices include computer, server & printer, communicate using protocol to exchange data, enabling applications like e-mail, file sharing & internet access.

★ Nodes :- Any device that can send, recieve or forward data within a network, such as computer, phones or printers.

★ Networking devices :- Devices that manage and support network fns., including routers, switches, hubs & access points.

★ Transmission media :- The physical or wireless medium through which data travels, such as ethernet cable or WiFi.

★ Protocols :- Sets of rules that govern how devices communicate & exchange data.

★ IP Address :- Unique identifiers for ~~dev~~ devices on a network, enabling routing & communication.

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NETWORK HARDWARE :~

Types of Networks

LAN
(Local Area)
- limited geogra. area.
- range [10m-1km]
- private network

WAN
(Wide Area)
- large geographical area.
- Eg:- Internet

PAN
(Personal area)
- very short range
- Range (1-9m)
- Eg:- Bluetooth hotspot.

MAN
(metropolitan area)
- connects devices across a city.
- range (10-50km)
- either private or public network.

WAN → MAN → LAN → PAN
~ decreasing order ~

NETWORK MODELS :~

➤ OSI Model [Open System Interconnection]:~
A conceptual model that divides network communication into 7 layers, each responsible for a specific set of functions.

➤ TCP/IP Model:~
A protocol model that simplifies network communication into four layers, widely used in the internet.

Application
Presentation
Station
Transportation
network
Data-link
Physical
} Layers.

A.L (Application)
(station + presentation layer)

Network hardware :~

- User machines in a network are called hosts.
- Transmission technology:-
  - → Point to point subnets - connect individual pairs of machines.
  - → Broadcast subnets - message is broadcast over a network & all machines have possibility of recieving the message.
    - Unicasting - one sender, one reciever
    - Broadcasting - recieved & processed by every machine on network.
    - Multicasting - recieved & processed by subset of machines.

connected by subnets. ↓ made up of transmission lines & switching elements.

Date

Expt. No. _____ Page No. _____

Personal area network [PAN] :-

PANs let devices communicate over the range of a person. [Eg. computers & peripherals].
→ Short range technologies use 'master-slave' paradigm.

Local Area network [LAN] :-

Used to connect personal computers to share resources.
→ LANs run at speeds of 100MBps to 1GBps. [Eg: Home office].
→ Various topologies are possible [bus-based, ring-based].
→ privately owned network that operates within & thereby a single building like a home, office or factory.
→ Newer LANs can operate upto 10GBps.

Metropolitan Area network [MAN] :-

→ Site is blw LAN & WAN, covers towns & cities. [high speed connectivity].
→ End-points spread over a city or a part of it, it may be public or private.
→ It supports data & voice.

Wide Area Network [WAN] :-

→ Covers countries or continents.
→ Two configurations :-
  ↳ Switched WAN - connects the end systems which has router.
  ↳ Point to point WAN - a line leased from a telephone or cable TV.

Protocol layering :-

Protocol defines the rules that both the sender & receiver and all intermediate devices need to follow to be able to communicate effectively.

- Advantages :-

It allows us to seperate devices from implementation.
We can use intermediate devices, to reduce cost of communication.

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★ Logical connections - represents that we have layer-to-layer communication.

★ **Layer :-**

Can be defined as a black box with inputs & outputs, without concern abt. how inputs are changed into outputs.

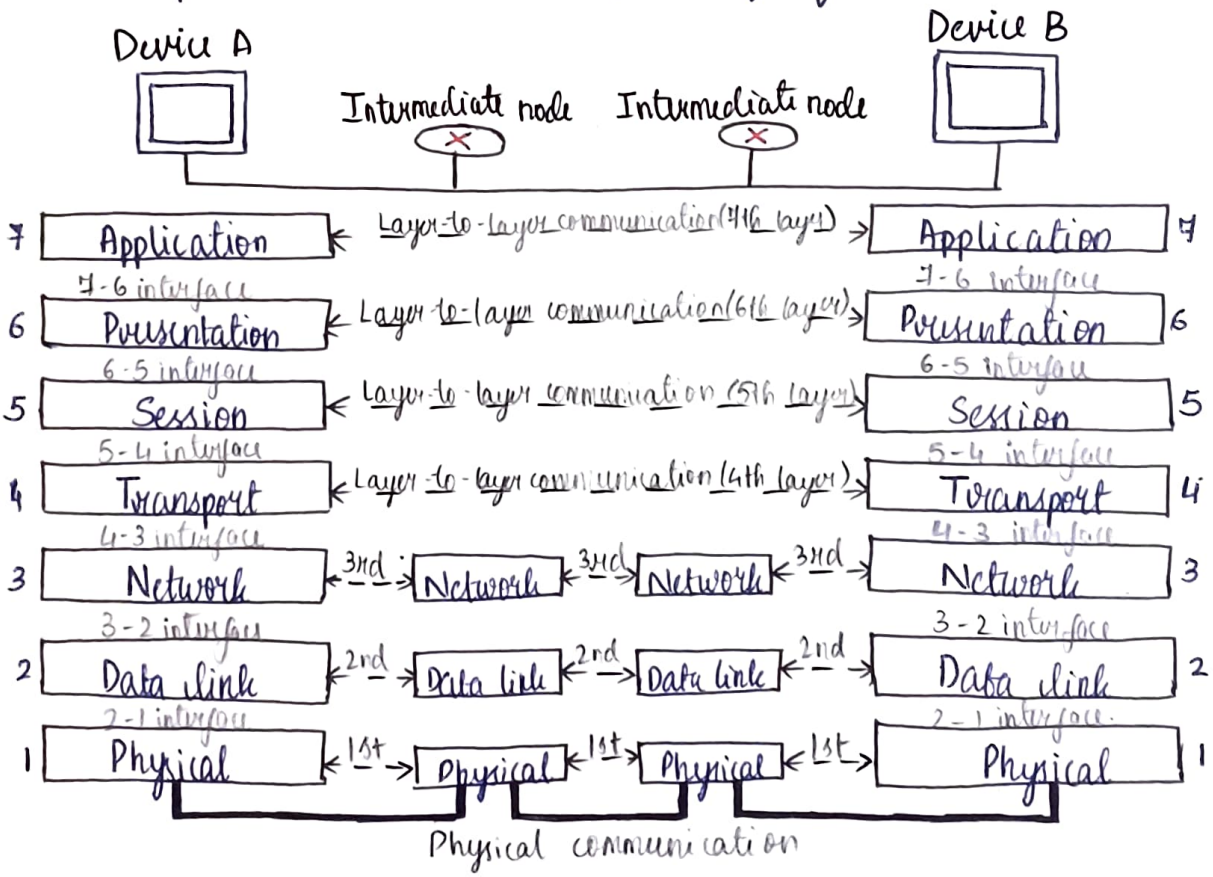
Principles :-

→ Bidirectional communication (need 2 end devices, to perform 2 opp. tasks).

→ Two objects under each layer at both sides should be identical.

➢ OSI LAYERS :-

- An open system is a set of protocols that allows any two different systems to communicate regardless of their underlying architecture.
- The purpose of OSI model is to show how to facilitate communication b/w different systems without requiring changes to the logic of underlying hardware and software.
- It is not a protocol, it is a model for designing a robust network arch.



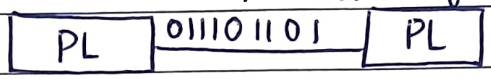
Date _____

Expt. No. _____ Page No. _____

- **Physical layer :-**

The physical layer coordinates the functions required to carry a bit stream over a physical medium.

→ It deals with the mechanical & electrical specifications of the interface and transmission media.



→ Transmission rate - the number of bits sent each second is also defined by the physical layer.

→ Line configuration - Point to point (2 devices are connected together through a dedicated line).

→ Physical topology

→ Transmission mode.

- **Datalink layer :-**

The data link layer transforms the physical layer, a raw transmission facility to a reliable link. It makes the physical layer appear error-free to the upper layer (network layer).

→ Framing - DLL divides the stream of bits received from the network layer into manageable data units called frames.

→ Physical addressing - If frames are to be distributed to different systems on the network, the DLL adds a header to the frame to define the sender and/or receiver of the frame.

→ Flow control - DLL imposes a flow control mechanism to prevent overwhelming line receiver.

→ Error control - It adds reliability to the physical layer.

★ **Access control :-**

When two or more devices are connected to the same link, DLL is necessary to determine which device has control over the link at a given time.

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- **Network layer :-**

The network layer is responsible for source-to-destination delivery of a packet, possibly across multiple networks (links). [IGMP protocol]

→ Logical addressing - The physical addressing implemented by the data link layer handles the addressing problem locally.

→ Routing - When independent networks/links are connected together to create internetworks or a large network, the connecting devices route or switch the packets to final destination.

- **Transport layer :-**

The transport layer is responsible for process-to-process delivery of the entire message. A process is an application program running on the host.

→ Service point addressing - delivery not only from one computer to another but also from one specific process to another.

→ Segmentation - A message is divided into transmittable segments with each segment having a serial number.

→ Can be connectionless or connection oriented.

→ Flow control is end-to-end rather than ~~across~~ across a single link.

→ Error control is process-to-process rather than across a single link.

- **Session layer :-**

The services provided by the first four layers are not sufficient for some processes.

→ Dialog control - The session layer allows 2 systems to enter into a dialog. It allows the communication b/w two processes either in half duplex or full duplex mode.

→ Synchronization - allows a process to add checkpoints into a stream of data.

- **Presentation layer :-**

It is concerned with the system & semantics of the information exchanged between two systems.

Responsibilities :-

→ Translation - exchange of info in the form of character strings, numbers etc.

→ Encryption - Conversion of original info to another form for safety purposes.

Date _____

Expt. No. _____ Page No. _____

→ Compression - reduces the number of bits contained in the information.

- **Application layer :~**

The application layer enables the user, whether human or software, to access the network. It provides user interfaces and support for services such as electronic mail, remote file access & transfer, shared database management & other types of distributed information services.

★ Servius provided by Appl. layer :-

→ Network Virtual terminal - It is a software version of a physical terminal and allows user to log on to a remote host.

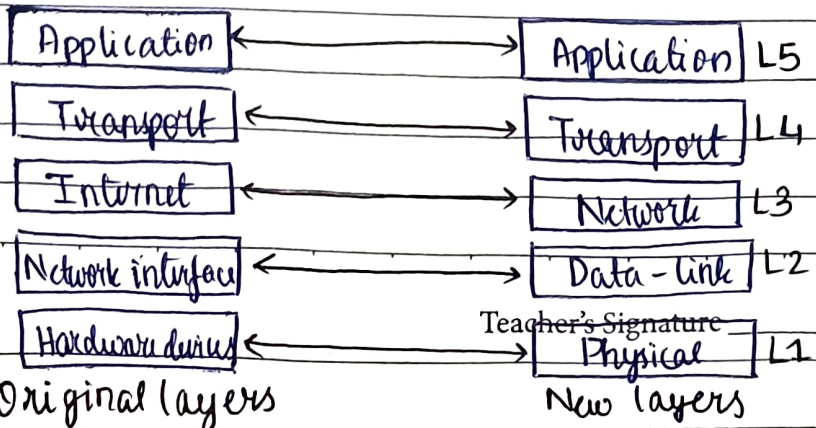
→ File transfer, access & management (FTAM) -
- Allows user to access file in a remote host
- to remove files from a remote computer for use in the local computer.
- manage/control files in remote computer locally.

→ Email services - basis for email forwarding & storage.

→ Directory services - provides distributed db sources & access for global info about various services.

TCP/IP PROTOCOL SUITE :~

- Defined as four-software layers built on the hardware [originally].
- Today, it is a five layer model.
- Set of protocols organised in different layers are being used in the internet today.
- Hierarchial protocols means the upper level are supported by one or more lower level protocols.



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★ Layered architecture :-

Each device is involved with a set of layers depending on the role of the device in the internet.

➤ Difference b/w TCP/IP & OSI layers :~

TCP/IP	OSI
• TCP/IP has 5 layers	• OSI has 7 layers
• TCP/IP is more reliable	• Less reliable
• Not very strict boundaries.	• Strict boundaries.
• Uses both session & presentation in application layer itself.	• Different session & presentation layers.
• Transport layer in TCP/IP does not provide assurance delivery of packets.	• Provides assurance.
• Network layer only provides connectionless services.	• Both connectionless & connection oriented services.

➤ Similarities b/w TCP/IP & OSI Layers :~

- Logical structure - both have layered architecture (easy troubleshooting and understanding)
- Network layer - present for routing packets.
- Transport layer - reliable data delivery.
- Application layer - for user-level interactions

NETWORK TOPOLOGIES :~

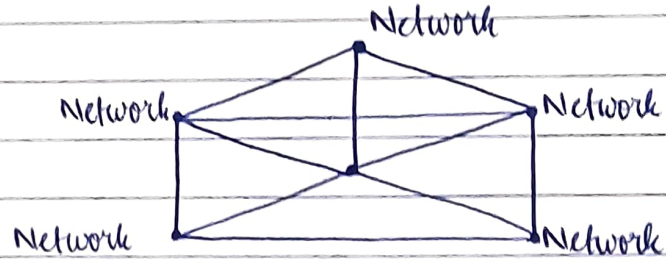
- Packet - A message ~~between~~ broken into smaller units for transmission on a network.
- Bus/Ring config - packet is sent off around ring on its own.
- Complete network - A station is directly connected to every other station on the network.
- Loop - Each ~~point is~~ packet is transmitted along the line until it encounters another computer.

Date

Expt. No. Page No.

MESH Topology :-

- To link n devices fully connected, mesh has: $n(n-1)/2$ physical channels - full duplex communication.
- Every device on the network has/must have n-1 ports.
- Example:-
  8 devices in mesh has links : $n(n-1)/2 = 8(7)/2 = 28$
  no. of links = 28
  no. of ports = $(n-1) = 7$.
- Every link is dedicated point-to-point link.
- The term dedicated means, the link only carries traffic b/w the two devices it connects.

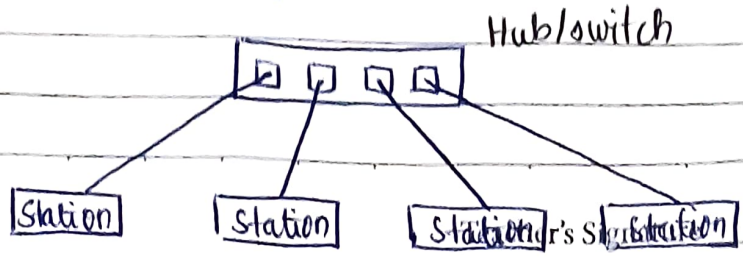


- Advantages:-
  → Each connection carries its own data load.
  → A mesh topology is robust.
  → Privacy or security
  → Fault identification & isolation.

- Disadvantages:-
  → Big amount of cabling
  → Big no. of I/O ports
  → Installation, reconnection are difficult.
  → Sheer bulk (wiring) > available space
  → Hardware connect could be expensive.

STAR Topology :-

- Dedicated point-to-point to a central controller (hub).
- No direct traffic b/w devices.
- The control acts as an exchange.

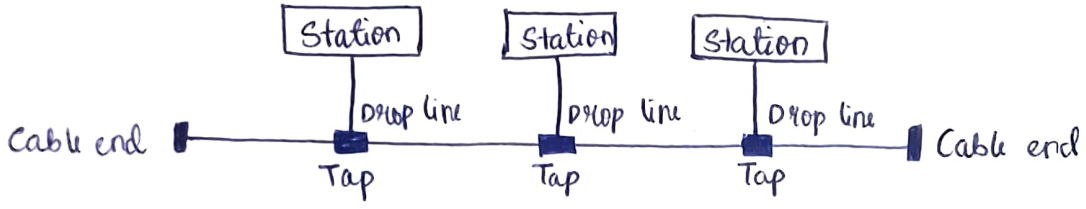


- Advantages :-
  - → less expensive than mesh
  - → Easy to install & reconfigure
  - → Less cabling
  - → Addition, moves & deletion required one connection.
  - → Robustness: One fail does not affect others.
  - → Easy fault identification & fault isolation.

- Disadvantages :-
  - → Dependency of whole topology on one single point (hub).
  - → More cabling than other topologies (ring or bus).

BUS Topology :-

- It is multipoint.
- One long cable acts as a backbone.
- Used in the design of early LANS, & Ethernet LANS.



- Advantages :-
  - → Ease of installation
  - → Less cables than mesh, star topo.

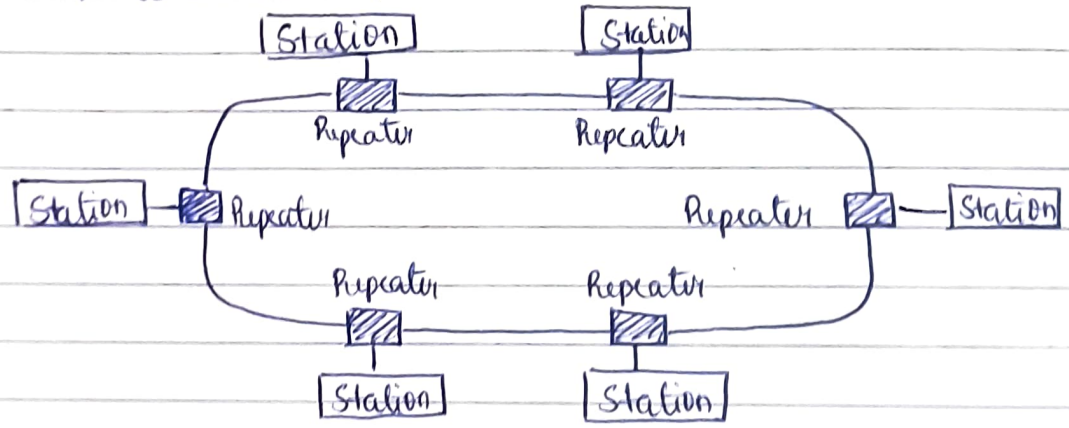
- Disadvantages :-
  - → Difficult reconnection & fault isolation.
  - → Adding new device requires modification of backbone.
  - → Fault or break stops all transmission.
  - → The damaged area reflects signals back in the direction of the origin, creating noise in both directions.

Date _____

Expt. No. _____ Page No. _____

RING Topology :~

- Each device has dedicated point-to-point connection with only two devices on either side of it.
- A signal is passed along the ring in one direction from device to device until it reaches its destination.



- **Advantages :-**
  - → Easy to install & reconfigure
  - → Connect to immediate neighbours
  - → Move two connections for any moving
  - → Easy fault isolation.
- **Disadvantages :-**
  - → Unidirectional
  - → One broken device can disable the entire network.

Hybrid Topology :~

Ex :- having a main star topology with each branch connecting several stations in bus topology.

→ It is difficult to incorporate mesh topology in hybrid topo. since it requires more cabling.

SWITCHING :~

- Switch - Device that creates temporary connections b/w devices.
- Internet is a switched network → switches connect links & forward data.

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- Types of switching networks-
  - → Circuit switching
  - → Packet switching (Datagram & Virtual-Circuit)
  - → ~~Mess~~ Message switching.

Circuit switching :-

- Dedicated path (circuit) b/w two stations.
- ~~Used~~ Uses FDM / TDM to divide link into channels.
- Phases: Setup → Data transfer → teardown.
- Data sent as continuous stream (not packets)
- Addressing (end-to-end) only in setup phase.
- Drawback: Inefficient - resources reserved for entire duration.
- Works at Physical layer.

Packet switching :-

- Data divided into packets
- Routers / switches use store-and-forward with queues
- Resources allocated on demand (no reservation)
- More efficient than circuit switching, but may cause delay.
- Datagram Networks :-
  - → Connectionless (no setup/teardown)
  - → Each packet treated independently
  - → Destination address stay same throughout.
  - → Routing via dynamic routing tables.
  - → Packets may arrive out of order.
  - → Switching in the internet is based on datagram approach.
- Virtual Circuit networks :-
  - → Hybrid of circuit & datagram switching.
  - → Packets follow same path.
  - → Phases: Setup → transfer → teardown.
  - → Uses VCI (Virtual circuit identifier) for local addressing.
  - → Delay: One time setup + teardown.
  - → Works at Data link layer.

Message switching :-

- Entire message sent as a unit with destination address.
- Uses store-and-forward at each intermediate node.
- No dedicated path.
- Provides dynamic routing based on available info.

Date _____

Expt. No. _____ Page No. _____

- Drawback : Higher delay due to storing full message at each node.

Comparison :~

- Circuit switching - Dedicated, continuous, inefficient but predictable.
- Datagram switching - Connectionless, dynamic, efficient, but may have delays/loss.
- Virtual circuit - Balance b/w the two, predictable path with packetization.
- Message switching - Entire message forwarded, high delay, not widely used today.

TRANSMISSION MEDIA :~

- Communication channel that carries information (bits) from sender to reciever via electromagnetic signals.
- Physical path b/w transmitter & reciever.
- Copper network - electrical signals
  Fiber network - light pulses
- Exists at Physical layer (OSI Model)
- Types : Wired (guided) & Wireless (unguided).

Factors affecting Transmission media :-

- Bandwidth - higher bandwidth → higher data transmission rate.
- Transmission impairment - recieved signal ≠ transmitted signal.
  Causes :- Attenuation - Signal energy loss with distance
  Distortion - Change in signal shape due to frequency components.
  Noise - Unwanted signal added during transmission.
- Interference - Disruption caused by unwanted signals.

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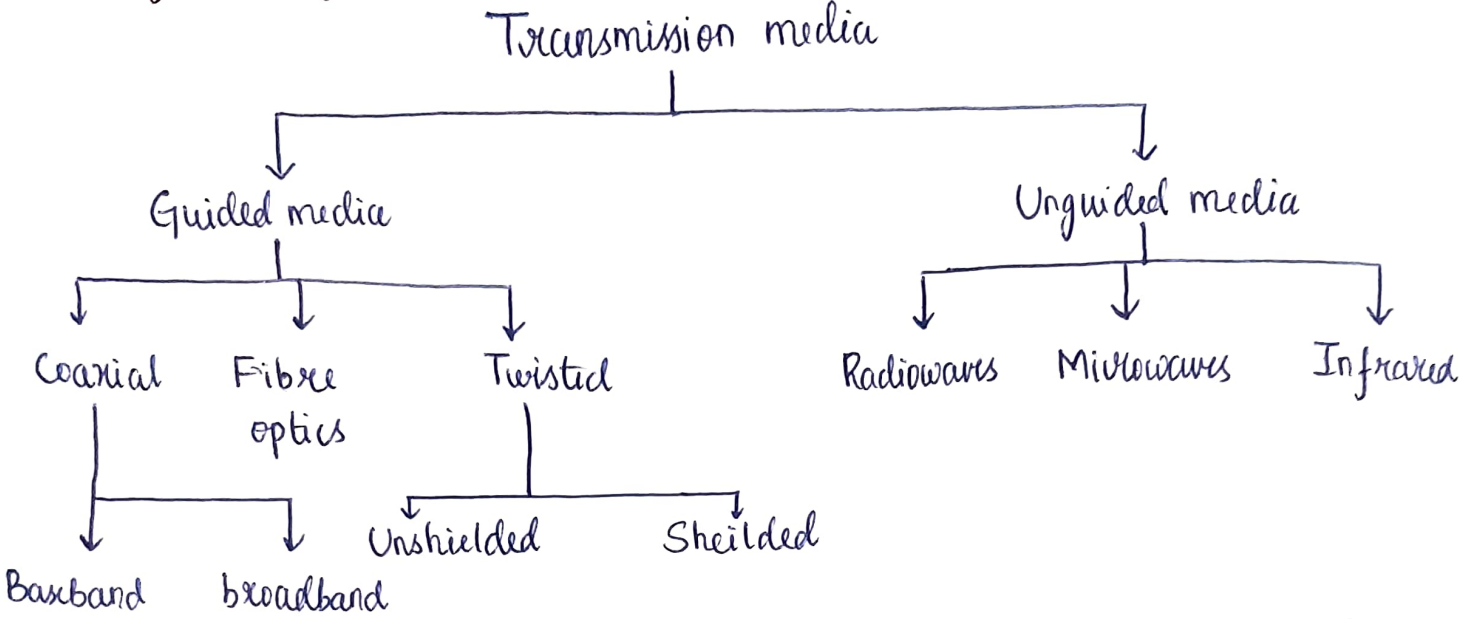
Classification of transmission media :-



- **Guided (Wired) Media :-**

  Signals transmitted through a physical medium.

  ⇒ **Coaxial cable :-**

  - **Structure :-**
    - Inner conductor (copper)
    - Non-conductive cover (insulation)
    - Outer conductor (copper mesh - protects from EMI)
  - **Types :-**
    - Baseband - single high speed signal
    - Broadband - multiple signals simultaneously.
  - Advantages - high speed, better shielding, higher bandwidth.
  - Disadvantages - Costly, single fault can collapse entire network.
  - Example : TV cables

  ⇒ **Fibre optic cable :-**

  - Uses light pulses instead of electricity.
  - **Structure :-**
    - Core (glass / plastic strand for data)
    - Cladding (glass layer)
    - Jacket (plastic protective coating)

Date _____

Expt. No. _____ Page No. _____

- Advantages:-
  - Higher bandwidth
  - Faster speed
  - Longer & distance transmission
  - Better reliability, immune to EMI
- Applications:- Internet backbone, medical imaging, telecommunication.

⇒ Twisted pair cable:-
- Two insulated copper wires twisted in spiral form.
- Cheap, lightweight, easy to install.
- Frequency range: 0-3.5 KHz.
- Unshielded twisted pair:-
  - Widely used in telecom & LANs.
  - Pros: Cheap, easy install, supports high-speed LAN.
  - Cons: Limited distance (attenuation).
- Shielded Twisted pair:-
  - Copper wires with shielding mesh.
  - Higher data rate, more reliable.
  - Pros: Higher capacity than UTP, less interference.
  - Cons: More expensive, higher attenuation than UTP/coaxial

- Unguided (Wireless) Media:-

Transmission via electromagnetic waves in air.

⇒ Radio waves:-
- Frequency: 3KHz - 1GHz.
- Omnidirectional (no alignment needed b/w antennas)
- Applications: FM radio, TV, cordless phones.
- Advantages:
  - Good for multicasting
  - Large coverage, penetrates walls
  - Higher transmission rate.

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⇒ Microwaves :-

- Frequency : 1 GHz - 1000 GHz
- Two types:
  - terrestrial microwave - ground-based antennas
  - Satellite microwave - communication via orbiting satellites.
- Characteristics:
  - Frequency ranges: 4-6 GHz, 21-23 GHz
  - Bandwidth : 1-10 Mbps
  - Short distance → inexpensive
  - long distance → costly (needs tall towers)
  - Affected by environment & antenna size.

⇒ Infrared :-

- Frequency : 300 GHz - 400 THz
- Short-range communication
- Applications: Remote controls, phone-to-phone transfer, PC-mobile transfer.
- Characteristics:
  - High bandwidth = high data rate
  - Cannot penetrate walls → isolated communication per room.

MIND MAP ~ UNIT 1

UNIT 1

- Network Hardware
  - Point to point
  - Broadcast subnets
    - Unicasting
    - Broadcasting
    - Multicasting
  - LAN — Local Area — 10m - 1km
  - WAN — Wide area
  - PAN — Personal area — 1-9m
  - MAN — metropolitan area — 10-50km
- Network Models
  - OSI Model
    - Physical layer
    - Data Link layer
      - Framing
      - Error control
      - Flow control
    - Network layer
      - Logical addressing
      - Routing
    - Transport layer
    - Session layer
    - Presentation layer
    - Application layer
  - TCP/IP Model
    - Application
    - Transport
    - Network
    - Data link
    - Physical
- Switching
  - Circuit switching
    - Works at physical layer
    - Setup - data transfer - teardown
  - Packet switching
    - Datagram — Connectionless
    - Virtual Circuit — Hybrid of circuit & datagram. Works at DLL
  - Message switching
    - Entire message sent as unit
    - No dedicated path.
- Network Topologies
  - MESH Topo. — n(n-1)/2
  - BUS Topo. — Multipoint — one long cable
  - STAR Topo. — Dedicated point to point central controller. — Hub/switch
  - RING Topo. — dedicated p-to-p connection with two devices.
  - Hybrid Topo.
- Transmission media
  - Exists at physical layer.
  - Factors affecting :-
    - Bandwidth
    - Interference
    - transmission impairment
      - Distortion
      - Attenuation
      - Noise
  - Guided media
    - Coaxial
      - Baseband
      - Broadband
    - Fibre optics
    - Twisted
      - Unshielded
      - Shielded
  - Unguided media
    - Infrared
    - Radio waves
    - Microwaves

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Expt. No. _____ Date _____ Page No. _____

UNIT-2

Syllabus :-

Data link layer :-

Design Issues, Framing, Error detection & correction, elementary data link protocols: simplex protocol, A simplex stop & wait protocol for an error free channel, A simplex stop and wait protocol for noisy channel. Sliding window protocols: A One-bit sliding window protocol, a protocol using Go-back-N, a protocol using selective repeat, Example data link protocols, medium access sub layer : The Channel allocation problem; Multiple access protocols : ALOHA, CSMA; Collision free protocols, Ethernet.

The data link layer (in OSI Model) lies between the physical layer & network layer.

- Converts the raw transmission facility of the Physical layer into a reliable, error-free link.

Design issues :-

- Providing services to the network layer
- Framing
- Error control
- Flow control.

SERVICES TO THE NETWORK LAYER :-

- Every layer uses the services of the one below & provides services to the one above.
- Data link layer uses the Physical layer and provides services to the network layer.

Types of services :-

➤ Unacknowledged connectionless service :-

→ Independent frames sent without acknowledgements

→ No logical connection established or released.

PTO →

Date _____

Expt. No. _____ Page No. _____

▶ Acknowledged connectionless service :-
→ Frames sent without a logical connection but each frame is acknowledged.
→ Sender knows if a frame arrived correctly.
→ If not received in time - retransmission.

▶ Acknowledged connection oriented service :-
→ Connection is ~~oriented~~ established before data transfer.
→ Frames are numbered.
→ Guarantees :-
- Every frame is received once
- Frames are delivered in right order.

FRAMING :-

'Encapsulation of network layer packets into frames for transmission.'

• Frame structure :-
- Header
- Payload (data packet)
- Trailer

• Types of framing :-
- Fixed size :-
  • Frame size fixed - length acts as delimiter
  • Drawback - Internal fragmentation if data < frame size
  • Solution - padding.
- Variable size - Start & end of each frame must be defined explicitly.

• Framing methods :-
→ Character count :-
  • Header field specifies number of characters in frame.
  • Receiver counts characters to detect frame end.
  • Problem : Errors in count corrupt whole frame.

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→ Flag bytes with byte stuffing :-
- Uses a flag byte to mark frame boundaries.
- If flag pattern occurs in data → escape (ESC) is inserted.

→ Flag bits with bit stuffing :-
- Flag pattern used to mark frame
- Sender inserts 0 after 5 consecutive 1s in data
- Receiver removes stuffed 0s
- Prevents confusion with frame boundary.

→ Physical layer coding violations :-
- Applicable where physical layer encoding has redundancy.
- Example: If 1 is encoded as high-low and 0 as low-high, an illegal combination can mark frame boundaries.

ERROR CONTROL :-
- Ensures error-free link.
- Functions :-
  - Detecting / correcting transmission errors
  - Sending acknowledgement frames
  - Retransmitting lost frames.
  - Identifying & discarding duplicate frames.

FLOW CONTROL :-
- Prevents fast sender from overwhelming a slow receiver.
- Without flow control, frames may be lost even if no errors occur.
- Ensures smooth transmission speed matching.

★ Errors in transmission :-
- During transmission, signals can be corrupted due to interference.
- Types of Errors :-
  → Single bit error - only one bit changes (0→1 & 1→0)
  → Burst error - 2 or more bits change.

Date _____

Expt. No. _____ Page No. _____

Redundancy :~

"The central concept in detecting or correcting errors is redundancy."

- Sender - adds redundant bits
- Reciever - checks & removes redundant bits.

⟹ Block coding :-

- Message divided into blocks of k bits = datawords.
- Add r redundant bits → block length = n = k + r → codeword.

ERROR DETECTION TECHNIQUES :~

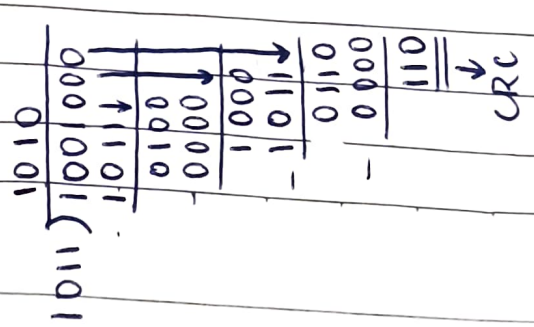
➤ Parity check code :-

- Linear block code
- k-bit dataword → n-bit codeword (n = k+1)
- Parity bit added :
  - → Even parity → total 1s = even
  - → Odd parity → total 1s = odd

➤ Cyclic Redundancy Check (CRC) :-

- Used in LANs & WANs.
- Steps :-
  - Find divisor length = n
  - Append (n-1) zeros to message
  - Perform binary division with divisor (generator polynomial) → remainder = CRC.
- Example :-

Dataword = 1001 → After appending ⇒ 1001 $\overbrace{000}^{(n-1)\ 0s}$; Divisor = 1011



Codeword = Data word + remainder

= 1001110 //.

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Checksum :-

- Message divided into m-bit units.
- Sender :-
  - → Divide into blocks
  - → Add all blocks - also add carry (if any)
  - → Take 1s complement → checksum.
- Reciever :-
  - → Add all blocks + checksum
  - → If result = all 0s → accepted, else rejected.
- Example :-

Original data -

1	2	3	4
10011001	11100010	00100100	10000100

Sender

```
  10011001
  11100010
 (1)01111011
    ↳      1
 ------------
  01111100
  00100100
 ------------
  10100000
  10000100
 ------------
 (1)00100100
    ↳      1
 ------------
  00100101
```

1s comp → 11011010 → checksum

Reciever

Sum + Checksum

```
  00100101
  11011010
 ------------
  11111111
```

complement = 00000000

Accept data.

ERROR CORRECTION TECHNIQUES :-

Hamming Code :-

- Detects and corrects single-bit errors.
- Uses redundant bits placed at positions that are powers of 2 (1, 2, 4, 8....)

Algorithm :-

- → Write bit positions in binary (1, 10, 11, 100...)
- → Positions = powers of 2 → parity bits
- → Rest → data bits
- → Each parity bit checks specific bit positions:
  - P1 → checks positions with LSB = 1 (1, 3, 5, 7)

Formula :-

$2^r \geq m + r + 1$

r = redundant bit

m = data bit

Date _____

Expt. No. _____ Page No. _____

- P2 → checks 2nd LSB = 1 (2, 3, 6, 7)
- P4 → checks 3rd LSB = 1 (4-7, 12-15 ...)
- P8 → checks 4th LSB = 1 (8-15, 24-31 ...)

• Example :-

11	10	9	8	7	6	5	4	3	2	1
1	0	0	r_8	1	1	0	r_4	1	r_2	r_1

m = 1001101

r_8	r_4	r_2	r_1	
0	0	0	1	- 1
0	0	1	0	- 2
0	0	1	1	- 3
0	1	0	0	- 4
0	1	0	1	- 5
0	1	1	0	- 6
0	1	1	1	- 7
1	0	0	0	- 8
1	0	0	1	- 9
1	0	1	0	- 10
1	0	1	1	- 11

r_1 = (1, 3, 5, 7, 9, 11)
⇒ r_1 10101 = 1
r_1 = 1

r_2 = (2, 3, 6, 7, 10, 11)
⇒ r_2 11101 = 0
r_2 = 0

r_4 = (4, 5, 6, 7)
⇒ r_4 011 = 0
r_4 = 0

r_8 = (8, 9, 10, 11)
⇒ r_8 001 = 1
r_8 = 1

11	10	9	8	7	6	5	4	3	2	1
1	0	0	1	1	1	0	0	1	0	1
			r_8				r_4		r_2	r_1

sender sends data to receiver

r_1 = (1, 3, 5, 7, 9, 11)
1 1 0 1 0 1 = 0

r_2 = (2, 3, 6, 7, 10, 11)
0 1 1 1 0 1 = 0

r_4 = (4, 5, 6, 7)
0 0 1 1 = 0

r_8 = (8, 9, 10, 11)
1 0 0 1 = 0

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DataLink Layer Protocols :~

Simple protocol :-

- Simplex communication.
- No flow or error control.
- Assumes reciever can always accept data.

Stop & Wait Protocol :-

- **Error free channel :-**
  - → Sender sends one frame - waits for acknowledgement.
  - → Reciever responds - half duplex channel sufficient.
- **Noisy channel :-**
  - → Uses CRC for error detection
  - → If frame corrupted, its discarded → retransmission needed.
  - → Window sizes : Sender = 1, Reciever = 1.

★ Piggybacking - ACKs are delayed & sent along with outgoing data frame to save bandwidth.

Sliding Window Protocol :-

- Allows multiple frames in transit.
- Ensures reliable and gradual delivery.
- Used in TCP.
- **Types :-**
  - → **Go-Back-N ARQ :-**
    - If error occurs, reciever discards erroneous + following frames.
    - Sender retransmits from error point.
  - → **Selective repeat ARQ :-**
    - Reciever stores correct frames, even if some lost.
    - Only retransmits ~~erroneous~~ erroneous/missing frames.

Date _____

Expt. No. _____ Page No. _____

Go-Back-n	Selective Repeat
• Go-back-n requires all retransmission of the succeeding frame along with the lost or damaged frame.	• In selective repeat, only the specific damaged or lost frame is retransmitted.
• All subsequent frames including the lost frames will be discarded.	• The frame that has been lost is the only one that is discarded.
• Window size is N.	• Window size is also N.
• Go-back-n protocol is less complex.	• Selective repeat is more comp.
• Neither sender nor receiver need sorting.	• Reciever side needs sorting to sort the frames.
• Type of Acknowledgement is cumulative.	• Type of Acknowledgement is Individual.
• Efficiency :- N/(1+2*a)	• Efficiency :- N/(1+2*a)

Protcols

- Noisless channel
  - → Simplex
  - → Stop & wait
- Noisy channel
  - → Stop & Wait
  - → Go-back-N
  - → Selective repeat.

▶ Protocols in practice :-

→ LANs - use data link layer protcols for local interconnection.

→ WANs - mostly point to point links.

- SONET (Synchronous Optical network) :-
  - → Optical fibre links in WANs
  - → Used to connect ISP routers.
- ADSL (Asymmetric digital subscriber line) :-
  - → Provides megabit/sec internet over telephone local loop.
  - → Millions of homes connected using ADSL.

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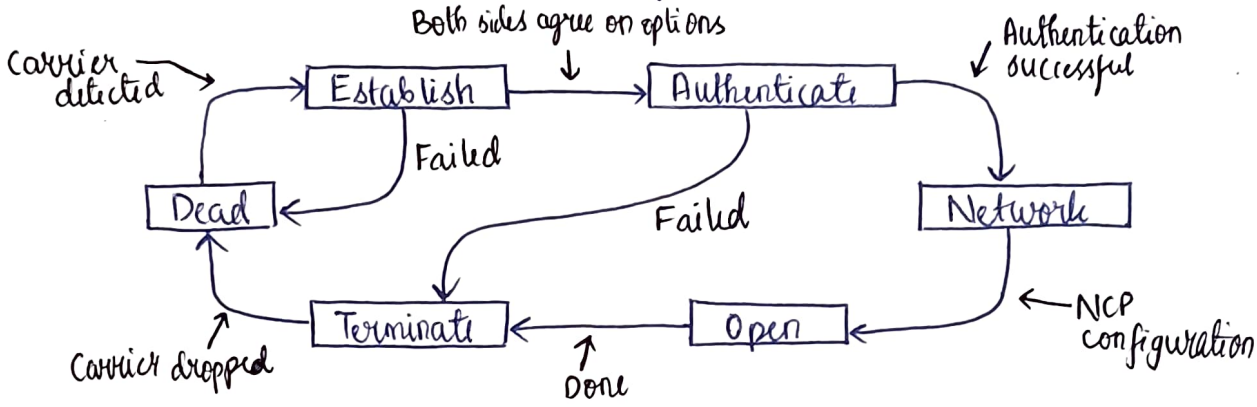
- **PPP (Point-to-Point protocol):-**
  - → Standard protocol for point-to-point links (SONET, ADSL, leased lines, modems)
  - → Improved version of SLIP.
  - → **PPP features:-**
    - Framing method - defines frame boundaries + error detection.
    - LCP (Link Control protocol) - sets up, tests, & tears down links.
    - NCP (Network Control protocol) - negotiates network layer options.

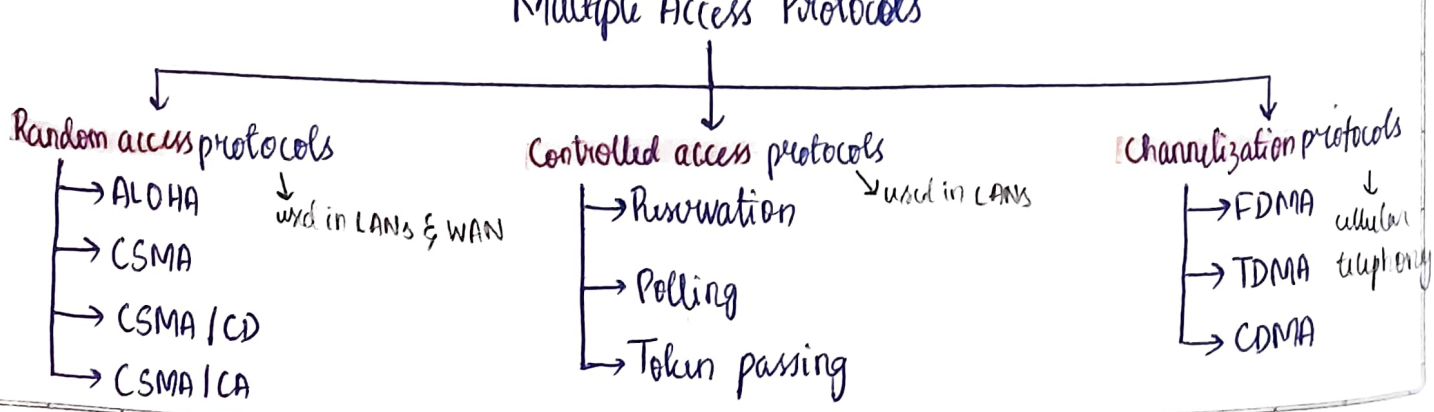


State diag. for bringing PPP link up & down.

→ High level data link control.

- → **PPP Vs HDLC:-**
  - PPP - byte oriented, ~~bit stuff~~ uses byte stuffing
    HDLC - bit oriented, uses bit stuffing.
  - HDLC → reliable transmission with sliding windows + ACKs
    PPP → framing, error detection, authentication, supports multiple protocols.

MULTIPLE ACCESS PROTOCOLS :-



Date _____

Expt. No. _____ Page No. _____

'When nodes / stations share a common link (multipoint / broadcast link) a multiple access ~~protocti~~ protocol is required to coordinate access.'

Random Access Protocols :-

- No station is superior or controls another
- No scheduled time for transmission → random access.
- Stations compete for medium access.
- **ALOHA :-**
  - → Designed for radio LAN
  - → Collision occurs when two stations transmit simultaneously - garbled data.
  - → **Pure ALOHA :-**
    - Station transmits whenever it has data.
    - Collisions possible, retransmission after random time.
    - Timeout = 2 x propogation delay ($2T_p$)
    - Retransmission max attempts = K_{max}.
  - → **Slotted ALOHA :-**
    - Time divided into slots (T_{fr})
    - Stations transmits only at start of slot.
    - Vulnerable time = T_{fr}.
- **CSMA [Carrier Sense multiple access] :-**
  - → Sense before transmit / Listen before talk.
  - → Reduces but cannot eliminate collisions.
  - → Vulnerable time = T_p
  - → **Persistence methods :-**
    - 1 - persistent - send immediately if idle (highest collision). Used in ethernet.
    - Non - persistent - wait random time if busy - less collision but less efficient.
    - p - persistent :-
      - Channel divided into time slots.
      - With probability p → transmit, with $q = 1 - p$ → wait for next slot.
      - Reduces collision, improved efficiency.

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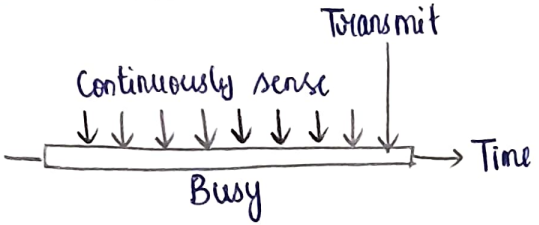
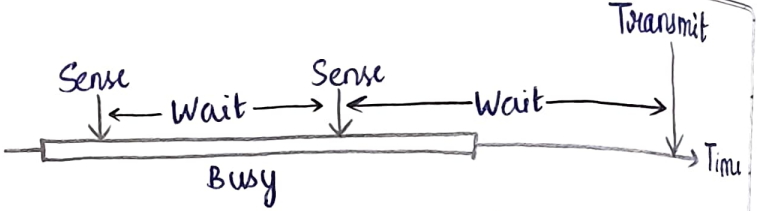
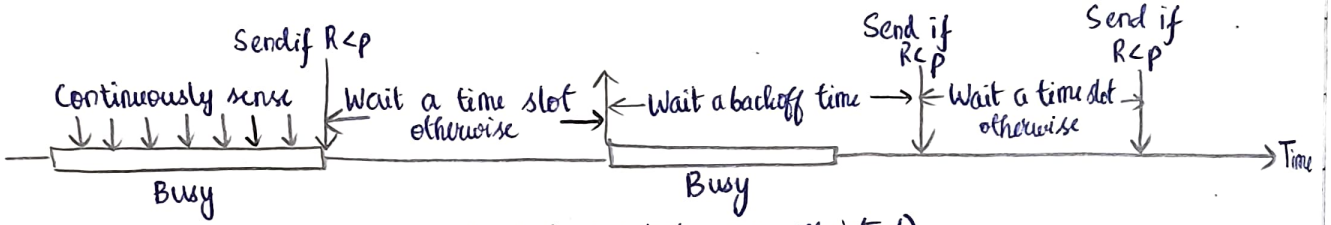


(a) 1-Persistent

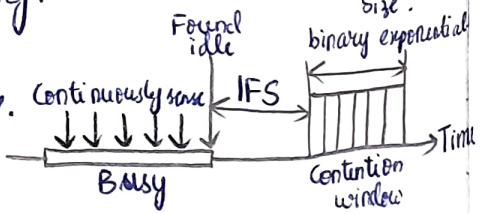


(b) Non persistent



(c) p-persistent (1 persistent + non-persistent)

- **CSMA/CD [Collision detection]:-**
  - → Monitors medium after sending
  - → If collision occurs - retransmit.
  - → Minimum frame size = $2T_p$ (must detect collision before finishing transmission)
  - → Differences from ALOHA:-
    - Persistence process added
    - Transmission + collision detection continuous.
    - Jamming signal sent to notify all stations.
- **CSMA/CA [Collision Avoidance]:-**
  - → Designed for wireless networks.
  - → Strategies: (→ when idle channel is found, the station does not send immediately.)
    - Interframe space (IFS):- wait time before sending.
    - Contention window :- random backoff slots
    - Acknowledgement (ACK) :- ensures correct delivery.



  - → Network Allocation vector (NAV):
    - RTS frame contains channel duration.
    - Other stations set NAV timers → avoid collision.

→ DIFS - DCF IFS - distributed coordination function IFS.
SIFS - short interframe space
RTS - request to send
CTS - clear to send

Controlled Access Protocols :-

Stations consult before sending → authorization required.

- **Reservation :-**
  - → Time divided into intervals.
    - → Reservation frame (with N minislots, one per station)
    - → Station marks its minislot → transmits in next interval.
- **Polling :-**
  - → Primary station controls communication.
  - → Poll = check if station has data.
  - → Select = ask secondary to receive.
  - → Models :-
    - Centralised polling - primary manages all
    - Distributed polling - no primary; stations follow polling order list.
  - → Drawback - if primary fails, entire system shuts down.
- **Token passing :-**
  - → Stations organised in ~~localised~~ logical ring.
  - → Access passed via token from predecessor → current → successor.
  - → Only token holder transmits.

Channelization Protocols :-

Channelization is a multiple access method in which the available bandwidth of a link is shared in time, frequency and through code, among diff. stations.

- **FDMA [Frequency division multiple access] :-**
  - → Bandwidth divided into frequency bands.
  - → Each station allocated one band
  - → Guard bands used to prevent overlap.
- **TDMA [Time division multiple access] :-**
  - → Channel divided into time slots.
  - → Each station gets a slot for transmission
  - → Synchronization needed for slot alignment.

CDMA [Code division multiple access] :-

- → All stations transmit simultaneously over one channel.
- → Each station uses a unique code to modulate its data.
- → Reciever uses same code to extract data.

ETHERNET :-

→ Also wired LAN.

Date _____

Expt. No. _____ Page No. _____

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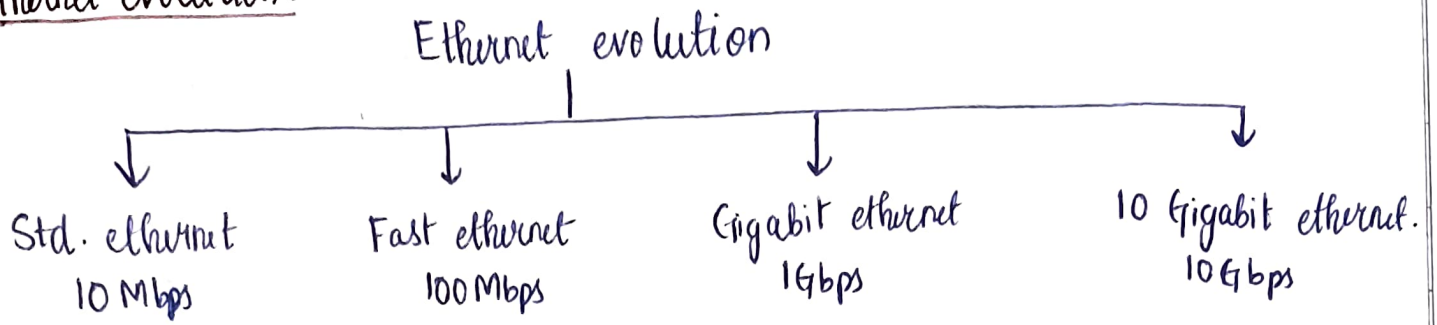
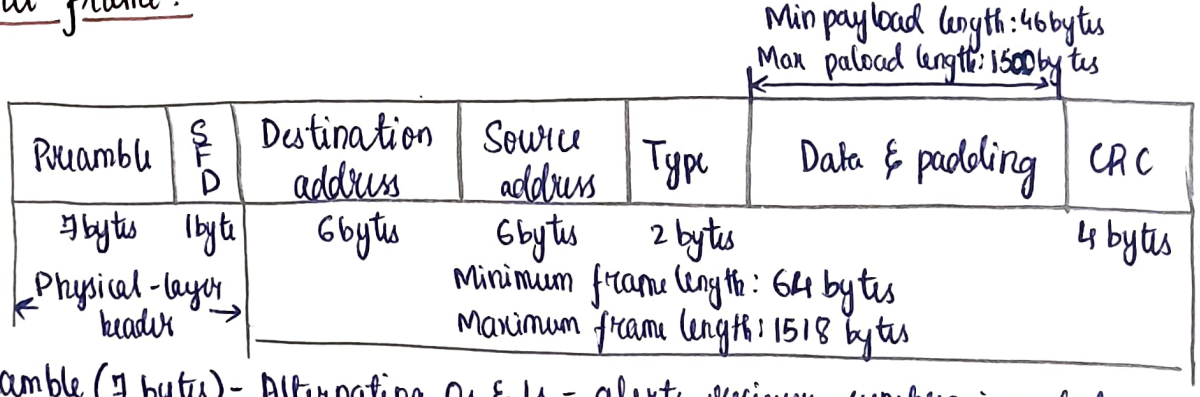
Ethernet evolution :-



★ Standard ethernet :-

- Data rate of 10 Mbps.
- Characteristics :-
  - → Connectionless & unreliable service. (each frame is independent of prev. & next frame; unreliable like IP & UDP.)
  - → if receiver finds about corruption, it silently drops the frame.
  - → Frame format
  - → Frame length.
  - → Ethernet frame :-



- Preamble (7 bytes) - Alternating 0s & 1s - alerts receiver, synchronizes clock.
- Start frame delimiter (SFD) (1 byte) - Marks start of frame.
- Destination address (6 bytes) - Receiver's MAC
- Source address (6 bytes) - Sender's MAC
- Type (2 bytes) - Upper layer protocol (IP, ARP, OSPF etc)
- Data (46 - 1500 bytes) - Encapsulated higher layer data.
- CRC (4 bytes) - Error detection (CRC-32)

⇒ Addressing :- Each device has Network Interface Card (NIC) with unique MAC address.

Ethernet implementations:-

- 10Base5 - Thick ethernet:-
  → First ethernet standard.

Expt. No. → Bus topology using thick coaxial cable. Page No. _____

→ Uses external transceiver connected via tap.
→ Max cable length : 500m

- 10Base2 - Thin ethernet:-
  → Bus topology with thinner, flexible coaxial cable.
  → Easier installation.
- 10BaseT - Twisted pair ethernet:-
  → Star topology
  → Each station connects to a hub with a twisted pair cable.
  → Max cable length : 100m
- 10Base F - fiber ethernet:-
  → Uses fiber-optic cables
  → Star topology with stations connected to hub.

★ Fast Ethernet:-

- Goals:-
  → Upgrade to 100Mbps.
  → Backward compatibility with standard ethernet.
  → Same 48 bit address & frame format.
- Feature - Autonegotiation:-
  → Allows devices to agree on data rate & mode
  → Supports backward compatibility
  → Enables device capability detection.

★ Gigabit Ethernet:-

- Goals:-
  → Upgrade to 1Gbps.
  → Same addressing, frame format, min/max length.
  → Support autonegotiation.

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- Modes:-
  - → Full duplex:-
    - Switch buffers traffic.
    - No collisions
    - Cable length limited by attenuation
  - → Half duplex:-
    - Uses CSMA/CD
    - Stations connected in star via hub/switch.
- Implementations:-
  - → Two wire: Fibre optic or STP.
  - → Four wire: Cat 5 twisted pair.

★ 10 Gigabit Ethernet:-

- Data rate: 10Gbps
- Supports LAN, MAN, WAN interconnection.
- Uses fibre optics only.
- Operates in full duplex only - CSMA/CD not used.
- Physical layer types - LAN PHY → for LANs
  WAN PHY → defines WAN compatibility

Comparison b/w wired & wireless LANs:-

	Wired LAN	Wireless LAN
Medium	Wires	Air
Hosts	host is ~~act~~ always connected to its network physically at a point.	host is not physically connected to the network.
Isolated LANs	A wired isolated LAN is a set of hosts connected via a link-layer switch.	A wireless isolated LAN, is a set of hosts that communicate freely with each other.
Connection to other networks.	It can be connected to another network such as internet using a router.	A wireless LAN may be connected to a wired network, or to another wireless LAN.

MIND MAP ~ UNIT 2

UNIT-2

Design Issues
- Services to network layer
  - Unack. connectionless
  - Ack. connectionless
  - Ack. connection oriented
- Framing
  - Character count
  - Bit stuffing
  - Byte stuffing
  - Physical layer coding violations.
- Error control
- Flow control.

Error detection techniques
- Parity check
- Cyclic redundancy check
- Chick sum

Error correction tech.
Hamming code

$2^r \geq m + r + 1$

- m = data bit
- r = redundant bit

DLL Protocols
- Noisy
  - Simplex
  - Stop & wait
- Noiseless
  - Stop & wait
  - Go back N
  - Selective repeat
- Protocols in prac.
  - SONET
  - ADSL
  - PPP

Multiple access proto.

Ethernet
- Wired LAN
  - Std. ether.
  - Fast ethernet
  - Gigabit ethernet
  - 10 Gigabit ethernet
- Ethernet implementations
  - 10Base5
  - 10Base2
  - 10BaseT
  - 10BaseF

Type → Type (2 byte)

Pream.	SFD	Dest. address	Source address	Data	Data	CRC

CRC (4 bytes)

- Preamble - 7 bytes
- SFD - Start frame delimiter (1 byte)
- Destination address (6 bytes)
- Source address (6 bytes)
- Data (46 + 1500 bytes)

Min frame length - 64 bytes
Max frame length - 1518 bytes.

Min payload length - 46 bytes
max payload length - 1500 bytes

Random Access protocols
- ALOHA
  - Pure
  - Slotted
- CSMA
  - Persistence
    - 1p
    - Pp
    - NP
- CSMA/CA
- CSMA/CD

Controlled access protocols
- Reservation
- Polling
  - Centralised
  - Distributed
- Token passing

Channelization
Bandwidth - time, frequency, code
- FDMA
- TDMA
- CDMA

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Expt No. _____

Date _____

Page No. _____

UNIT - 3

Syllabus :-

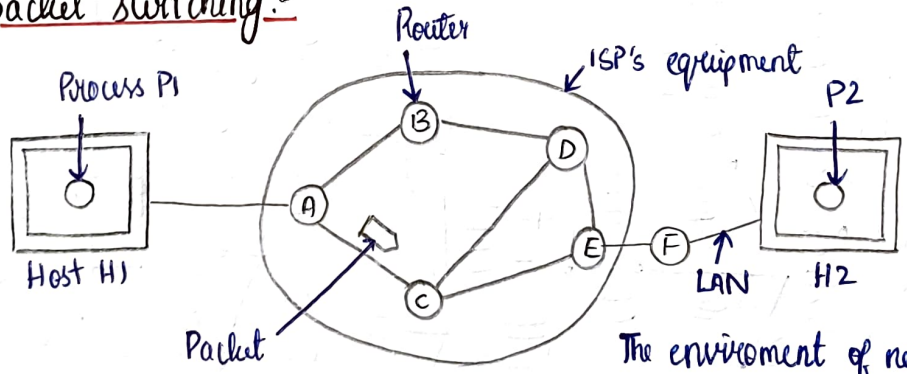
Network layer :-

Design issues, routing algorithms : The Optimality principle, shortest path algorithm, flooding, distance vector routing, link state routing, OSPF, BGP, Congestion control algo., Qual. of service, Internetworking, The network layer in the internet - The IP Ver. 4 Proto., Fragmentation & IP addresses, CIDR notation, ~~IPv6~~ IPv6, Int. control protocols, Basics of IP Suppost Protocols, Network Address Translation.

★ Network layer design issues :-

- Store and forward packet switching
- Services provided to transport layer.
- Implementation of connectionless service.
- Implementation of connection oriented service
- Comparison of virtual circuit and datagram networks.

➤ Store & forward packet switching :-



The environment of network layer protocols

- A host with a packet to send transmits it to the ~~the~~ nearest router, either on its own LAN or over a point-to-point link to the ISP.
- The packet is stored there until it has fully arrived and the link has finished its processing by verifying the checksum. Then it is forwarded to the next router along the path until it reaches the destination host, where it is delivered.

Date _____

Expt. No. _____ Page No. _____

Services provided to the Transport layer :-

- Services independent of router technology.
- Transport layer shielded from number, type, topology of routers.
- Network addresses available to transport layer use uniform numbering plan, even across LANs & WANs.

Implementation of connectionless service :-

- If connectionless service is offered, packets are injected into the network individually and routed independently of each other.
- No adv. setup needed.
- The packets are frequently called datagrams and network is called datagram network.
- If connection-oriented service is used, a path from source router all the way to the destination router must be established before any data packets can be sent.

Implementation of connection oriented service :-

- When a connection is established, a route from the source machine to the destination machine is chosen as part of the connection setup and stored in routers.
- Have a fixed path.
- If one connection is terminated, then VC (Virtual circuit) is also terminated. [assigned a number - (virtual circuit) VC number]

★ Comparison of Virtual Circuit & Datagram networks :-

Issue	Datagram network	Virtual Circuit network
Circuit setup	Not needed	Required
Addressing	Each packet contains the full source & destination address.	Each packet contains a short VC number.

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State information	Routers do not hold state information about connections.	Each VC requires router table space per connection.
Routing.	Each packet is routed independently.	Route chosen when when VC is setup; all packets follow it.
Effect of router failures	None, except for packets, lost during the crash.	All the VCs that passed through the failed router are terminated
Quality of service	Difficult	Easy if enough resources can be allocated in advance for each VC
Congestion control	Difficult	Easy if enough resources can be allocated in advance for each VC

⇒ **Routing algorithms:-**

- Optimality principle
- Shortest path algorithm
- Flooding ★ - 2m
- Distance vector routing
- Link state routing
- Routing in ad hoc networks
- Broadcast routing
- Multicast routing
- Anycast routing
- Routing for mobile hosts

Non-Adaptive (static)
→ Do not use measurements
→ static routes are downloaded at boot time.

Adaptive algorithms.
→ Change routes dynamically.
→ Gather information at runtime.

➤ **Properties of network :-**

- Correctness - The network should deliver the data to the correct destination, without mistakes.
- Simplicity - The network design & protocols should be easy to understand & manage.
- Robustness - The network should keep working even if some parts fail.
- Stability - The network should remain steady and not fluctuate too much in performance.
- Fairness - If two people are streaming videos, both should get decent quality, not just one person.

Date _____

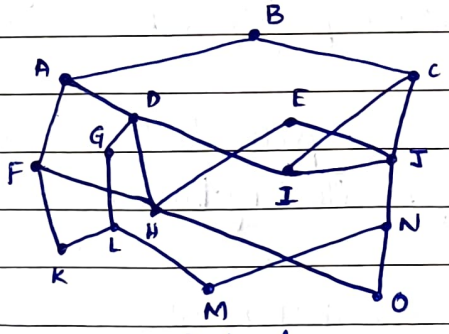
Expt. No. _____ Page No. _____

- Efficiency - The network should make the best use of resources.
  Ex:- sending data without wasting time or using too much energy.

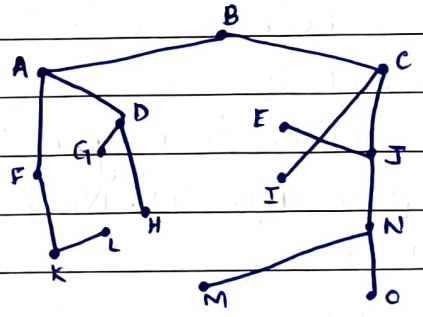
---CT 1---

Optimality principle:-

- If router j is on the optimal path from i to k, then the optimal path from j to k also falls along the same route.
- The purpose of any routing algorithm is to decide which O/P line an incoming packet should go.
- The optimal path from a particular router to another may be the least cost path, least distance path, least time path, least hops path or a combination of all.
- Sink Trees:-
  → The set of all optimal routes to a particular node forms a sink tree.
  → Sink trees are not necessarily unique.



- Subnet -



- Sink tree -

Shortest path routing:-

- Given a network topology and a set of weights describing the cost to send data across each link in the network.
- Find the shortest path from a specified source to all other destinations in the network.
- Shortest path algorithm first developed by E.W. Dijkstra.

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- Process:-
  → Mark the source node as permanent.
  → Designate the source node as the working node.
  → Set the tentative distance to all other nodes to infinity.
  → While some nodes are not marked permanent:-
    - Compute tentative distance from the source to all adjacent nodes.
    - Examine all nodes. Select node with smallest value & designate the node as permanent.

Flooding :-

- In flooding every incoming packet is sent out on every outgoing line except the one it arrived on.
- Flooding generates vast numbers of duplicate packets.
- In flooding, without control, a packet could keep ~~bou~~ ~~borrowing~~ bouncing around forever causing network overload. So we add a hop count limit.
- The source node attaches a sequence number to the packet. Every node keeps a record of packets it has already seen.
  - If a node recieves a new packet, forwards it to neighbors.
  - If a node ~~rec~~ recieves a duplicate packet, discards it.
- Advantages:-
  - Guaranteed delivery: As long as there's a path, the packet will reach the ~~dst~~dest.
  - Simple to implement: No need for complex routing tables.
- Disadvantages:-
  - Too many copies: The same packet travels many times, wastes bandwidth.
  - Network congestions: Can overload the network with duplicate packets.
  - Not efficient: A lot of unnecessary transmissions.
- It is robust.
- Effective in broadcasting information.
- It can be used as a metric.
- Practical in few settings:
  - Military applications
  - Distributed databases
  - Metric for comparison.

Date _____

Expt. No. _____ Page No. _____

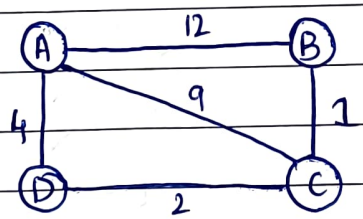
Distance vector routing :-

- A distance vector routing algorithm operates by having each router maintain a table (i.e. a vector) giving the best known distance to each destination and which link to use to get there.
- The distance vector routing algorithm is sometimes called - 'Bellman-Ford routing algorithm'.
- It was the original ARPANET routing algorithm & was also used in the internet under the name RIP (routing information protocol).
- It is iterative, distributed & asynchronous.
- It is a dynamic algorithm.
- Each router maintains a distance table known as 'vector'.
- Let $d_x(y)$ be the least cost path from node x to node y. The least costs are related by Bellman ford equation :-

$$d_x(y) = \min_v \{ c(x,v) + d_v(y) \}$$

(x: source; y: destination; v: intermediate node)

Example :-



Step 1 :- (direct to destination route.)

A

Dest.	Dist.	Next R.
A	0	A
B	12	B
C	9	C
D	4	D

B

Dest.	Dist.	Next R.
A	12	A
B	0	B
C	1	C
D	∞	–

(no direct route (hence ∞).)

C

Dest.	Dist.	NR
A	9	A
B	1	B
C	0	C
D	2	D

D

Dest.	Dist.	NR
A	4	A
B	∞	–
C	2	C
D	0	D

PTO →

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→ shortest path

Step 2:-

A

Dest.	Dist.	NR
A	0	A
B	7	DC
C	6	D
D	4	D

B

Dest.	Dist.	NR
A	7	DC
B	0	B
C	1	C
D	3	C

C

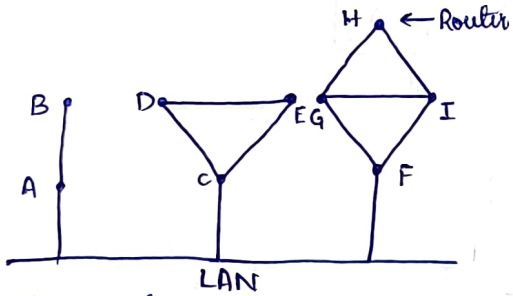
Dest.	Dist.	NR
A	6	D
B	1	B
C	0	C
D	2	D

D

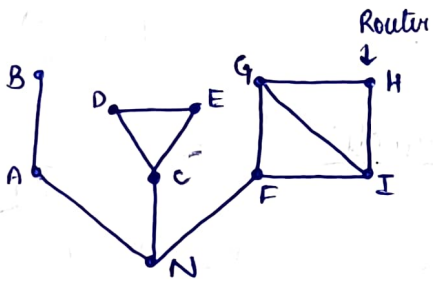
Dest.	Dist.	NR
A	4	A
B	3	C
C	2	C
D	0	D

Link state routing :-

- Type of routing algorithm where each router builds a complete map of the network by exchanging "link-state packets" with every other router.
- Five steps :-
  - Discover your neighbors and learn their addresses.
  - Measure the cost to each neighbor.
  - Construct a packet containing all this information.
  - Send this packet to all other routers.
  - Compute the shortest path to every other router.



- Nine routers & broadcast LAN -



- A graph model -

OSPF :-

The Open Shortest Path First Protocol (OSPF) is an intra-domain routing protocol based on link state routing. Its domain is also an autonomous system.

Types of Links

- Point-to-Point
- Transient
- Stub
- Virtual

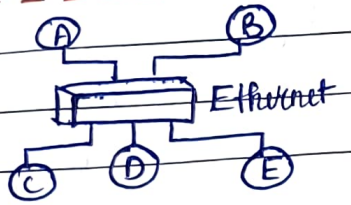
Date _____

Expt. No. _____ Page No. _____

⇒ Point - to - Point link :-

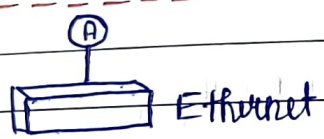
Direct exchange of messages blw two nodes within a network.

⇒ Transient network :-



Temporary state of links during a network topology change, which can cause forwarding loops until the network converges.

⇒ Stub link :-

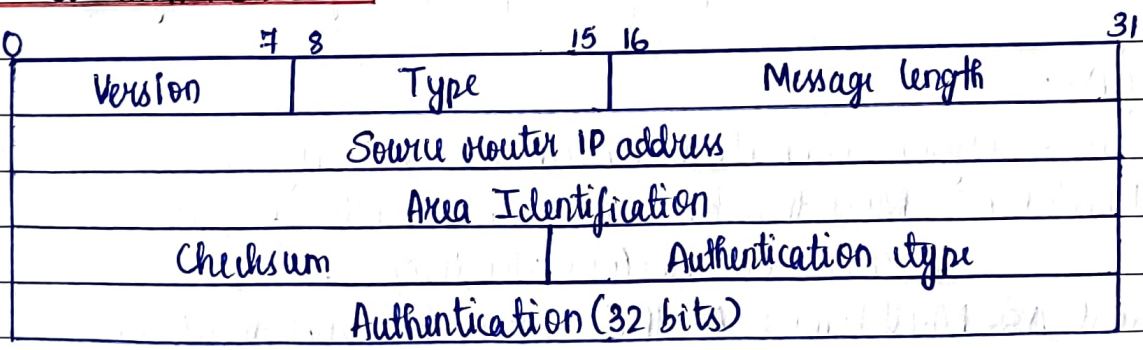


Refers to a link to a stub area with LSA.
does not accept external routes & uses default routes.

⇒ Virtual link :-

If due to some problem, the connectivity blw a backbone & an area is broken, a virtual link ~~must be~~ blw routers must be created by the administration to allow continuity of functions of the backbone as the primary area.

➤ OSPF Common header :-



This header contains all the necessary information to determine whether the packet should be accepted for further processing.

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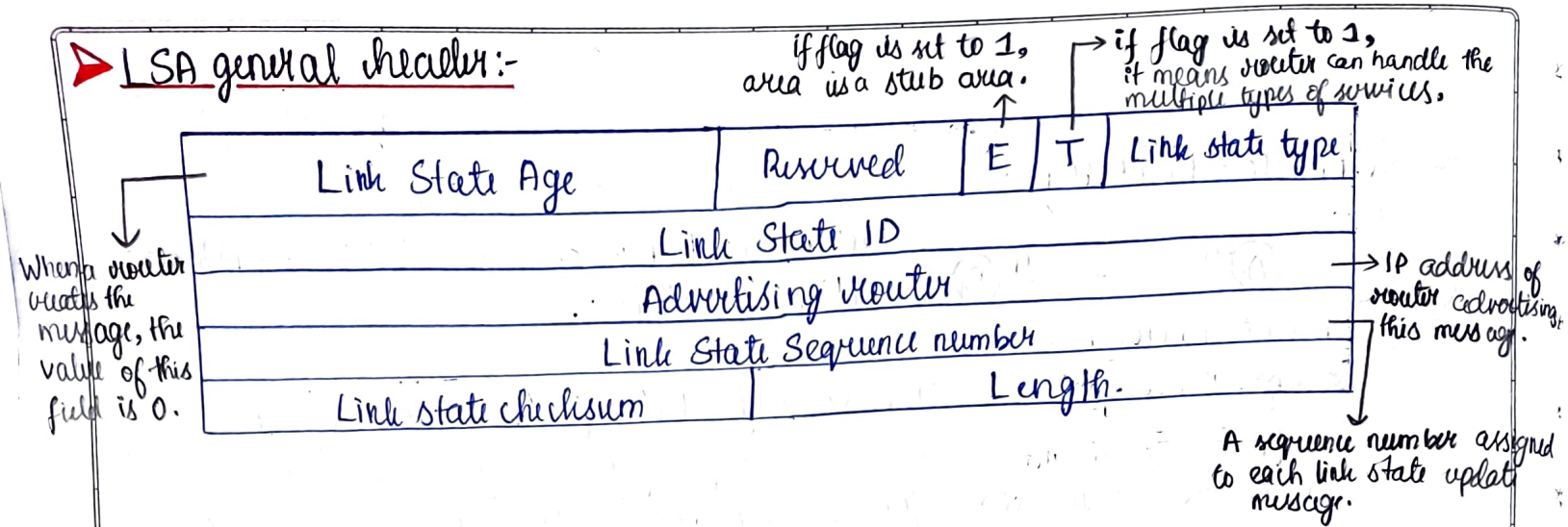
LSA general header:-



BGP :-

Border Gateway Protocol (BGP) is an interdomain routing protocol using path vector routing.

Path Vector Routing :-

- Sharing - A speaker in an AS shares its table with immediate neighbors.
- Updating - Adding the nodes that are not originally in its routing table.
- Loop prevention - A router checks to see if its AS is in the path list to destination.
- Policy routing - If one of the ASs listed in the path is against its policy, it can ignore the path & the destination.
- Optimum path - The optimum path is the path that fits the organization.

Types of AS :-

- Stub AS - only one connection to another AS (only a source or sink for data traffic)
- Multihomed AS - More than one connection to other AS, but it is still only a source or sink for data traffic.
- Transit AS - Multihomed AS that allows transient traffic.

★ NLRI :-

'Network layer reachability information.'

→ It defines the network that is actually advertised by this message.

→ Length field & IP address prefix.

→ BGP4 supports classless addressing and CIDR. → Classless Inter-Domain Routing.

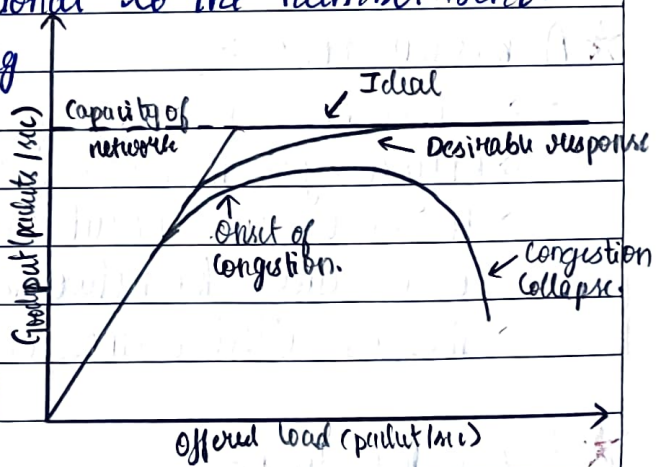
Date _____

Expt. No. _____ Page No. _____

NOTE:-
→ BGP uses the service of TCP on port 179.
→ BGP supports classless addressing and CIDR.

CONGESTION CONTROL:-

- Too many packets present in the network causes packet delay & loss that degrades performance. This is called congestion.
- The network & transport layer share the responsibility for handling congestion.
- The most effective way to control congestion is to reduce the load that the transport layer is placing on the network. This requires transport & network layers to work together.
- The number of packets delivered is proportional to the number sent.
- As the offered load approaches the carrying capacity, ~~bursts of~~ bursts of traffic occasionally fill up the buffers inside the routers and some packets are lost.
- The network is now congested.
- It may experience 'congestion collapse', in which performance plummets as the offered load increases beyond the capacity.



- 'goodput' - rate at which useful packets are delivered by the network.
- Approaches to Congestion control:-
  - Network provisioning
  - Traffic aware routing
  - Add Admission control
  - Traffic throttling
  - Loadshedding

↓ Slower (Preventative)
↓ Faster (Reactive)

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→ Eg:- turning on spare routers or enabling lines that are normally used only as backups.

★ Network provisioning :-

→ If there is a low bandwidth link on the path along which most traffic is directed, traffic is likely.

→ More often, links and routers that are regularly heavily utilized are upgraded at the earliest opportunity. This is called 'provisioning' & happens on a scale of months.

Eg:- routers may be changed to shift traffic away from heavily used paths by changing shortest path weights.

★ Traffic aware routing :-

→ To make the most of the existing network capacity, routes can be tailored to traffic patterns that change during the day as network users wake and sleep in different time zones.

→ Local radio stations have helicopters flying around their cities to report on road congestion, so that users can reroute their packets.

→ This is called traffic aware routing.

→ Splitting traffic across multiple paths is also helpful.

★ Admission Control :-

→ Sometimes it is not possible to increase the capacity.

→ Only way then, is to decrease the load.

→ In a virtual circuit network, new connections can be refused if they would cause the network to become congested.

→ This is called admission-control.

★ Traffic throttling :-

→ In a network, when congestion is approaching, it should tell the senders of the packets to slow them down.

→ Can be used in Virtual circuit & data-gram networks.

★ Load shedding :-

→ When the routers are inundated by packets that they cannot handle, they just throw them away.

Date

Expt. No. Page No.

QUALITY OF SERVICE :-

- An easy solution to provide good quality of service is to build a network with enough capacity for whatever traffic will be thrown at it. - 'overprovisioning'.
- The resulting network will carry application traffic without significant loss and, assuming a decent routing scheme, will deliver packets with low latency.
- Four issues must be addressed to ensure QoS:-
  - → What applications need from a network.
  - → How to regulate the traffic that enters the network.
  - → How to reserve resources at routers to guarantee performance.
  - → Whether the network can safely accept more traffic.
- Categories of QoS:-
  - → Constant bit rate. Eg:- telephony
  - → Real time variable bit rate. Eg:- Compressed videoconferencing.
  - → Non real-time variable bit rate. Eg:- Watching a movie on demand.
  - → Available bit rate. Eg:- File transfer.

Application requirements:-

- → A stream of packets from a source to destination is called flow.
- → A flow might be all the packets of a connection in a connection-oriented network; or all the packets sent from one process to another process in a connection-less network.
- → Four primary parameters:
  - bandwidth
  - jitter
  - reliability
  - delay

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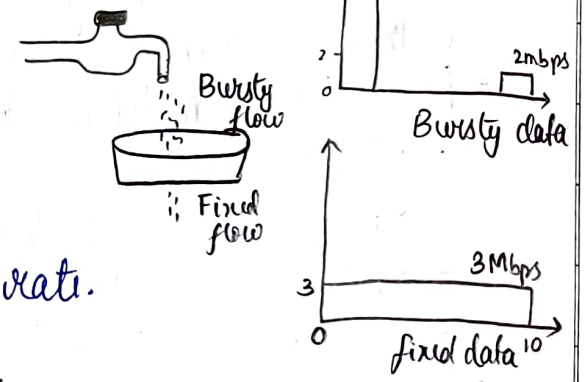
Traffic shaping :-

→ Traffic shaping is a technique for regulating the average rate and burstiness of a flow of data that enters the network.
→ When a flow is set up, the user & the network agree on a certain traffic pattern, for that flow.
→ This agreement is called 'Service Level agreement' (SLA).
→ Monitering a traffic flow is called 'traffic policing'.
→ Leaky bucket syndrome:-
  → Let us imagine a bucket with a small hole at the bottom where the rate at which ~~not~~ water is poured into the bucket is not constant & can vary but it leaks from the bucket at a constant rate.



Packet Scheduling :-

→ Types :- FIFO queuing
Priority queuing
Weighted fair queuing.

→ Algorithms that allocate router resources among the packets of a flow and between competing flows are called packet scheduling algorithms.
→ Three kinds of resources reserved for different flows :-
  - Bandwidth
  - Buffer space
  - CPU cycles.

Admission Control :-

→ A user submits a flow with specific QoS requirement to the network.
→ The network checks its capacity and existing commitments to decide acceptance or rejection.
→ If accepted, the network reserves capacity at routers to ensure QoS for the new flow.

Integrated Services :-

→ It was aimed at both unicast & multicast applications.
→ This protocol is used for making the reservations, other protocols are used for sending the data.

Date _____

Expt. No. _____ Page No. _____

→ RSVP allows multiple senders to transmit to multiple groups of recievers.
→ Permits individual recievers to switch channels freely, and optimizes bandwidth while at the same time eliminating congestion.

IP ADDRESSING

- IP Address is an address having information about how to reach a specific host, especially outside the LAN. → divided into two parts - host address, network address
- 32 bit unique address with address space of 2^{32}.
- Dotted decimal notation:-

10000000 00001011 00000011 00011111

128 . 11 . 3 . 31

- Hexadecimal notation:-

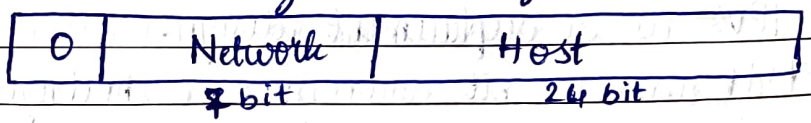
01110101 00011101 10010101 11101010

75 1D 95 EA

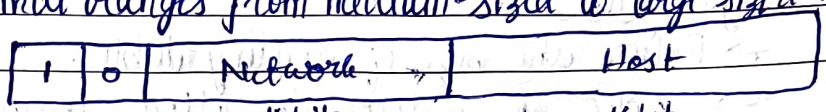
0x751D95EA

- Classes:-

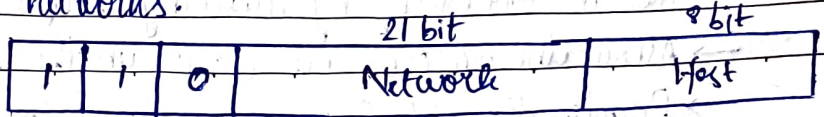
→ Class A:- [0-127]
- Assigned to networks that contain large number of hosts.
- Network ID is 8 bits
  Host ID is 24 bits



→ Class B:- [127-191]
- Assigned to networks that ranges from medium-sized to large-sized nets.
- Network ID is 16 bits
  Host ID is 16 bits



→ Class C:- [192-223]
- Assigned to small-sized networks.
- Network ID is 24 bits
  Host ID is 8 bits.



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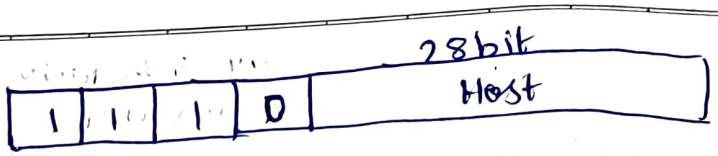
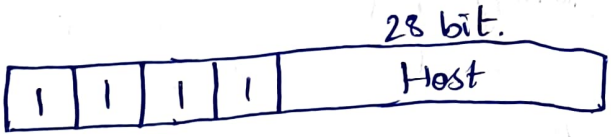
→ Class D :- [224-239]



- Reserved for multicasting
- Does not possess any sub-net mask.

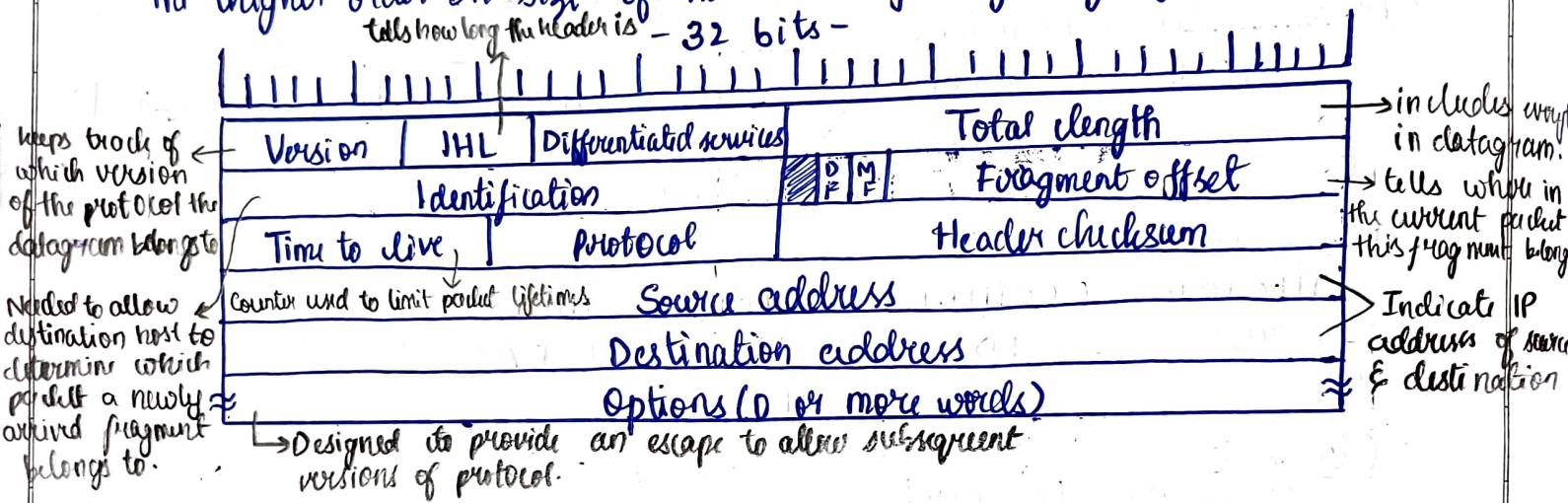
→ Class E :- [240 - 255]



- Reserved for experimental purposes.
- Does not have any sub-net mask.

IPv4 Protocol :-

- An IPv4 datagram consists of a header part and a body or payload part.
- The bits are transmitted from left to right and top to bottom, with the higher order bit size of the Version field ~~goint~~ going first.

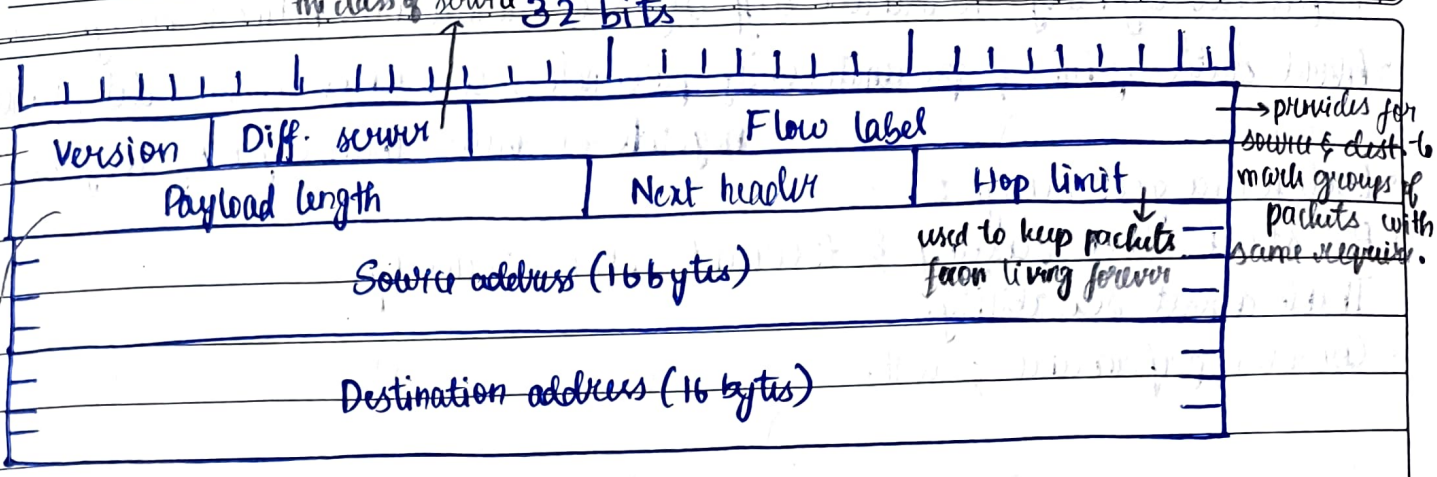


IPv6 Protocol :-

- IPv6 is a replacement design.
- It uses 128-bit addresses; a shortage of these addresses is not likely any time in the foreseeable future.
- Major goals :-
  - → Reduce the size of routing tables.
  - → Simplify the protocol, to allow routers to process packets faster.
  - → Provide better security.
  - → Aid multitasking by allowing scopes to be specified.
  - → Allow the protocol to evolve in the future.
  - → Permit the old & new protocols to coexist for years.

Date

Expt. No. Page No.



Differences b/w IPV4 & IPV6 :-

IPV4	IPV6
• 32-bit address length.	• 128-bit address length.
• Supports manual & DHCP address configuration.	• Supports Auto and renumbering address configuration.
• End-to-End connection integrity is unachievable.	• End-to-End connection integrity is achievable.
• It can generate 4.29×10^9 address space.	• Can produce 3.4×10^{38} address space.
• Security feature is dependent on application.	• IPSEC is inbuilt security feature.
• Address representation is decimal.	• Address representation is hexadecimal.
• Checksum field is available.	• Checksum field is not available.
• Has header of 20-60 bytes.	• Has header of 40 bytes fixed.

★ Network layer in the Internet :-

-The top 10 principles-

- Make sure it works : Do not finalise design until multiple prototypes & successful.
- Keep it simple : When in doubt, use the simplest solution.
- Make clear choices : Choose one out of multiple choices.

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- Exploit modularity: This principle leads directly to the idea of having protocol stacks.
- Expect heterogeneity: To handle diff. types of hardware; network design must be simple.
- Avoid static options & parameters: If parameters are unavoidable, let sender & receiver negotiate a value than defining a fixed value.
- Look for a good design
- Be strict when sending & tolerant when receiving.
- Think about scalability.
- Consider performance & cost.

PROTOCOLS :-

ARP [Address Resolution Protocol] :-

- Used to associate an IP address with MAC address.
- ARP is used to find the MAC address of the node when an internet address is known.
- Working :-
  - → When the host wants to know the physical address of another host on its network, it sends an ARP query packet that includes the IP address & broadcasts it over the network.
  - → Every host receives the packet; but only the desired host recognizes the IP address & sends back the physical address.

RARP [Reverse Address Resolution Protocol] :-

- If host wants to know its IP address, it broadcasts the RARP query packet that contains its physical address to the entire network.
- A RARP server recognizes the RARP packet & responds with host's IP address.
- Message format of RARP protocol is similar to ARP protocol.
- RARP frame is sent from one machine to another encapsulated in the data portion of a frame.

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Date _____

Expt. No. _____ → Messages used by DHCP:- Discover, Offer, Request, Decline, Release, Ack, Nack. Page No. _____

DHCP [Dynamic Host Configuration Protocol]:-

- Lets network administrators manage centrally and automate the assignment of IP addresses in an organisation's network.
- Without DHCP, an IP address must be manually entered at each computer.
- If computers move to another location, a new IP address must be entered.
- DHCP lets a network administrator supervise & distribute IP addresses from a central point & assign new IP addresses when comp. is plugged into a different place in the network.
- IP Address Allocation Mechanism:-
  - → Automatic Allocation: assigns a permanent IP address to host.
  - → Dynamic Allocation: assigns IP address to host for a limited period of time.
  - → Manual Allocation: Assigned by NA, and DHCP is used to ~~conv~~ convey the assigned address to host.

ICMP [Internet Control Message Protocol]:-

- Works with IP to send error messages and operational information, helping to diagnose and troubleshoot network problems like destination unreachable or excessive network traffic.

Type (8 bit)	Code (8 bit)	Checksum (8 bit)
Extended Header (32 bit)		
Data/Payload (Variable length)		

– ICMP Packet Format –

- → Type (8 bit):- Provides brief description of the message, so that the reciever knows how to respond to it.
- → Code (8 bit):- This field carries additional information about error message & type.
- → Checksum (16 bit):- Used to check the no. of bits of the entire message.
- → Extended header:- Points out the problem in the IP message.
- → Data/Payload:- IPv4 - 576 bytes; IPv6 - 1280 bytes.

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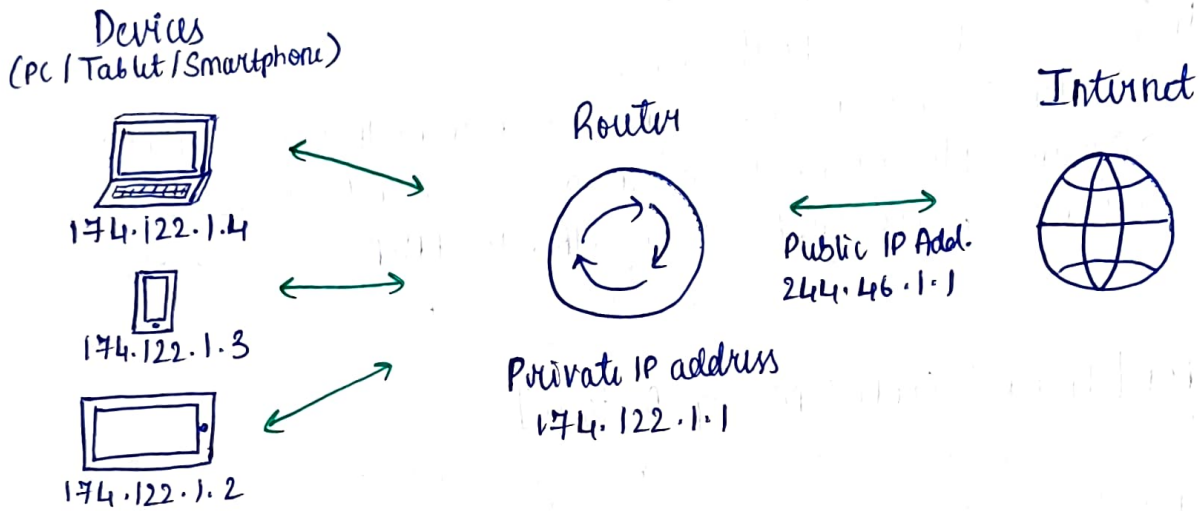
NET. ADD. TRANSLATION :-

- Network Address translation (NAT) is a process in which one or more local IP addresses are translated into one or more Global IP addresses.
- This is done to provide Internet access to local hosts.
- It makes the corresponding entries of IP address and port number in the NAT table.
- NAT generally operates on a router or firewall.



- Working:-
  Recieves the packet → Translates the address → Records the translation → Forwards the Packet
  Forwards the response ← Packet

- Benefits:-
  → IP Address Conservation: Crucial for conserving the limited supply of public IP Addresses.
  → Enhanced security: NAT provides extra security layer, making it more difficult for external devices to directly access internal devices.

- Types of NAT:-
  → Static NAT: One-on-one mapping of a private IP address to a singh public IP address.
  → Dynamic NAT: Maps private IP addresses to a pool of available IP addresses.
  → Port Address translation (PAT): Allows many devices to share a singh IP address with each internal connection using a unique port number to distinguish b/w them.

MIND MAP ~ UNIT 3

UNIT-3

Network layer design issues
- Store & forward packet swit.
- Services to TL
- Connectionless service
- Connection oriented service
- Diff. b/w VC & datagram network.

Routing algorithms
- Optimality principle
- Shortest path routing
- Flooding
- Distance vector rout.
- Link state routing

Props of network
- Correctness
- Simplicity
- Robustness
- Fairness
- Stability

OSPF
- Point to point
- Transient
- Stub
- Virtual

OSPF common header

LSA general header

BGP
Types of AS
- Stub
- Multihomed
- Transit

Path Vector Routing

Congestion Control
- Loop prevention
- Shaving
- Updating optimum path
- Policy routing

Load shedding

Network Provisioning

Traffic aware routing

Admission control

Traffic throttling

Quality of Service
Application requirements
- bandwidth
- jitter
- reliability
- delay

Traffic Shaping
- Leaky bucket

Packet Scheduling
- Bandwidth
- buffer space
- CPU cycles

Admission control

Integrated services

IP addressing
Classes
- Class A - 0-127
- Class B - 128-191
- Class C - 192-223
- Class D - 224-239
- Class E - 239-255

IPv4 Protocol

IPv6 Protocol

Network layer in the internet

Differences b/w IPv4 & IPv6

Network Address Translation
Devices ↔ Router ↔ Internet

Working

Benefits

Types
- Static NAT
- Dynamic NAT
- Post address translation (PAT)

Protocols
- ARP
- RARP
- DHCP
- ICMP

Expt. No. _____

Date _____

Page No. _____

Teacher's Signature _____

UNIT-4

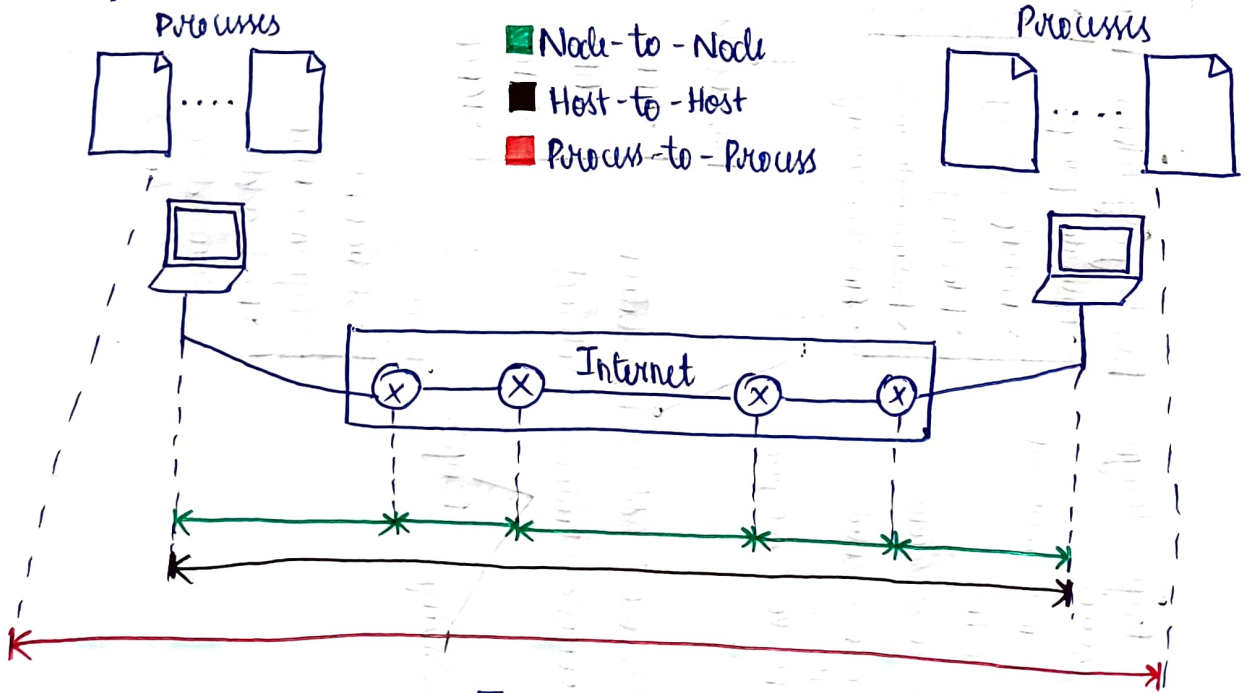
Syllabus:-

Transport Layer:-

Transport Services-Transport service primitives, Berkeley sockets, TCP/UDP Sockets, Elements of Transport protocols, The Internet Transport protocols: UDP, TCP - Intro. to UDP, Real time transport protocols, Introduction to TCP, The TCP service model, The TCP protocol, TCP Segment header, TCP Connection Establishment, TCP connection release, TCP Sliding Window, TCP Timer management, TCP Flow control, Congestion Control.

PROCESS-TO-PROCESS COMM.

- DLL is responsible for delivery of frames b/w two neighbouring nodes over a link. ↗ node-to-node delivery
- NL is responsible for delivery of datagrams b/w hosts. → host to host delivery.
- TL is responsible for process-to-process delivery - the delivery of a packet, part of a message; from one process to another.



- Types of data deliveries -

Date _____

Page No. _____

Expt. No. _____

Client / Server Paradigm :-

- A process on the local host, called a client, needs services from remote host called as server. Both processes have same name.
- Example :- To get the day & time from a remote machine,
  - we need a DayTime client process running on the local host.
  - we need a DayTime server process running on a remote machine.

Addressing :-

- Whenever we need to deliver something to a destination, we need an address.
- At the DLL, we need a MAC Address to choose one node among several nodes if the connection is not point-to-point.
- A frame in the DLL needs a destination MAC address for delivery & a source address for the next node's reply.

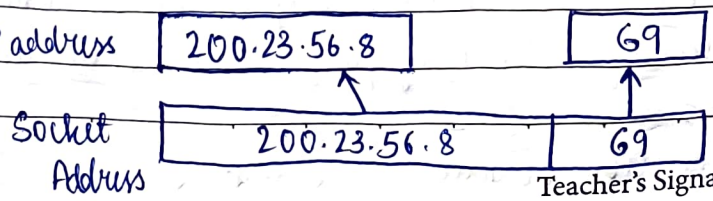
IANA Ranges :-

The IANA (Internet Assigned Number Authority) has divided port numbers into:-

- Well known ports :- 0-1023 ; assigned & controlled by IANA. registered to prevent duplication
- Registered ports :- 1024-49151 ; not assigned or controlled by IANA.
- Dynamic ports :- 49152-65535 ; neither controlled nor registered. → can be used by any process.

Socket Addresses :-

- The combination of an IP address and a port number is called a 'socket address'.
- The client socket address defines the client process uniquely just as the server socket address defines the server process uniquely.
- UDP or TCP contains the port numbers.



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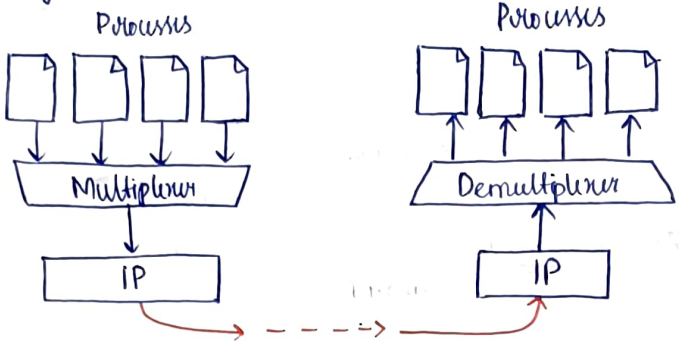
Multiplexing & Demultiplexing :-

- **Multiplexing :-**
  At the sender site, there may be several processes that need to send packets. However, there is only one transport layer protocol at any time. This is a many-to-one relationship and requires multiplexing.
- **Demultiplexing :-**
  At the receiver site, the relationship ~~to~~ is one-to-many and requires demultiplexing.



- Multiplexing and Demultiplexing -

Connectionless service :-

The packets are sent from one party to another with no need for connection establishment or connection release.

→ Packets are not numbered.

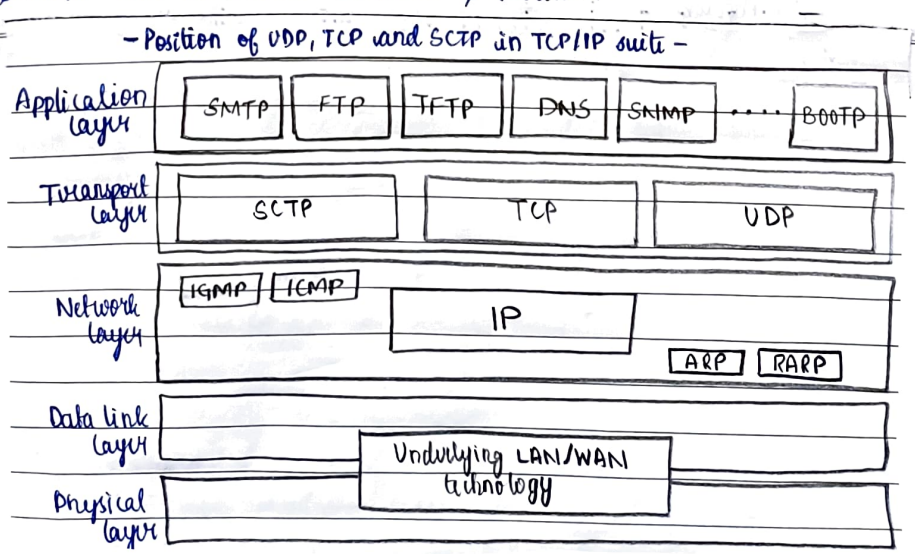
→ There is no acknowledgement either.

Connection-oriented service :-

A connection is first established b/w sender & receiver. Data is transferred. At the end connection is released.

Reliable & Unreliable service :-

- The TL service can be reliable or unreliable. If it needs reliability, we use a reliable transport layer protocol by implementing flow & error control at transport layer.
- UDP is connectionless & unreliable.
- TCP & SCTP are connection oriented & reliable.



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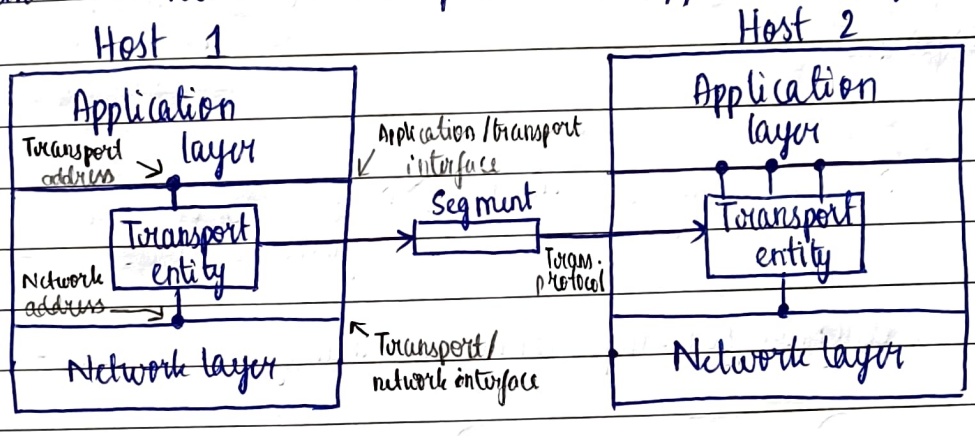
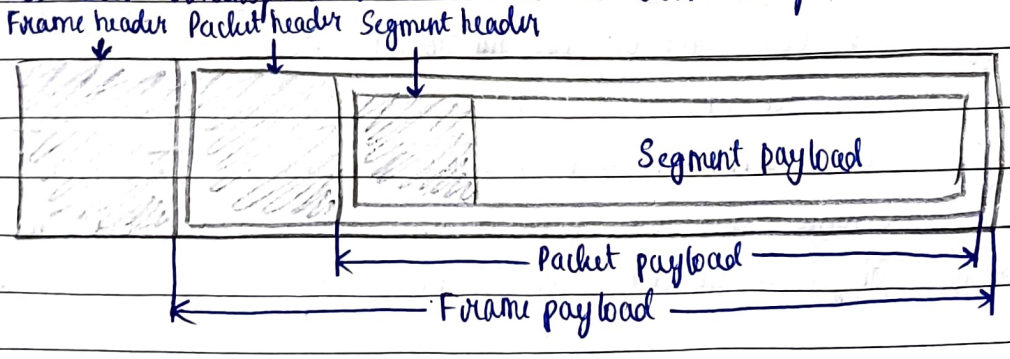
TRANSPORT SERVICES :-

- Goal :- Providing efficient, reliable & cost-effective data transmission to application layer processes.
- Uses services provided by the network layer.
- Logical link b/w network, transport and application layers.



Transport Service primitives :-

To allow users to access the transport service, the transport layer must provide some operations to application programs, that is, a transport service interface. Each transport service has its own interface.



Berkeley sockets :-

Sockets first released as Berkeley UNIX 4.2BSD software distribution in 1983. The primitives are now widely used for Internet programming on many operating systems, especially UNIX based systems.

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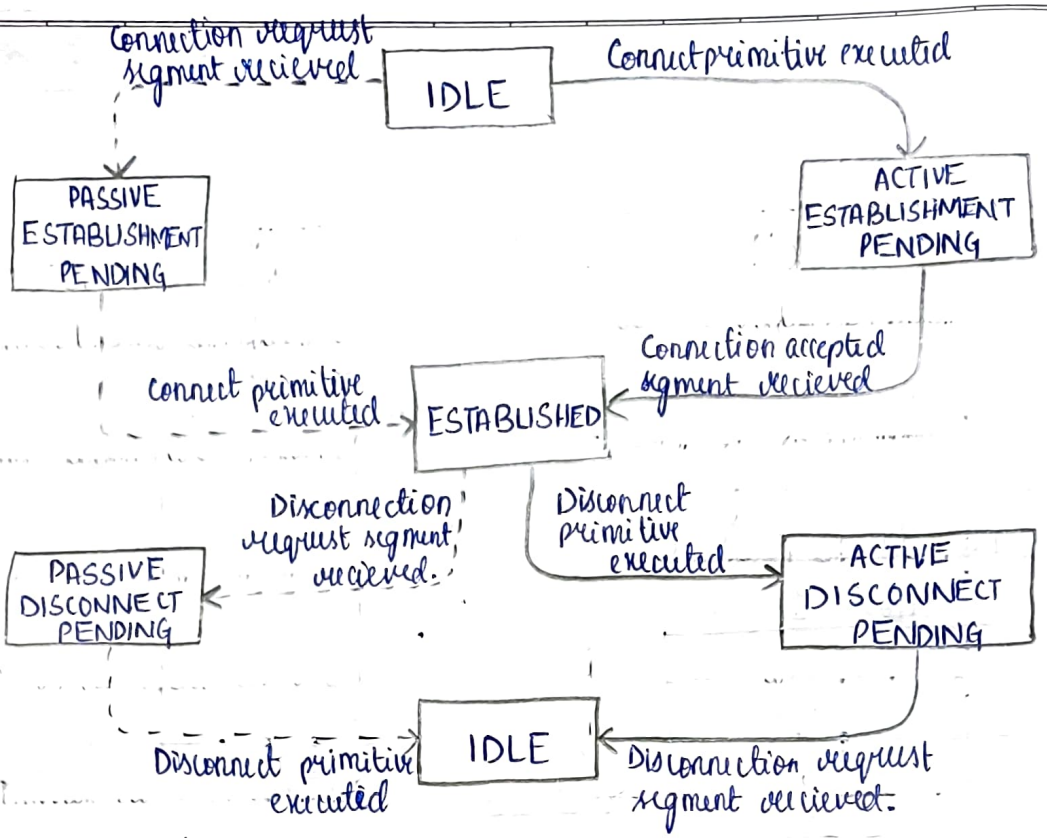
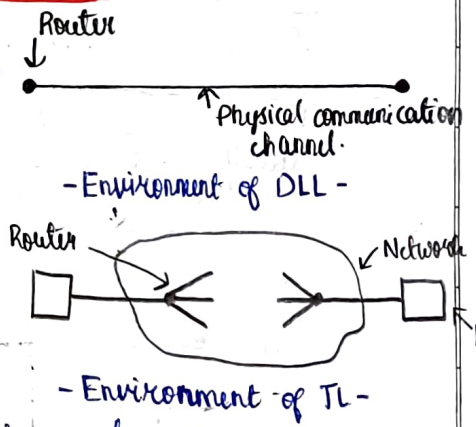


-State diagram for simple connection management scheme-

ELEMENTS OF TRANS. PROTOCOLS :-



→ The transport service is implemented by a transport protocol used b/w the two transport entities.

→ There are differences also, due to dissimilarities b/w the environments in which the two operate.

→ In DLL, two routers communicate directly via a physical channel, wether wired or wireless.

→ At the TL, the physical channel is replaced by entire network.

➢ Addressing:-

→ In internet, the endpoints are called ports.

→ The end point in the transport layer is called 'Transport Service Access point' (TSAP)

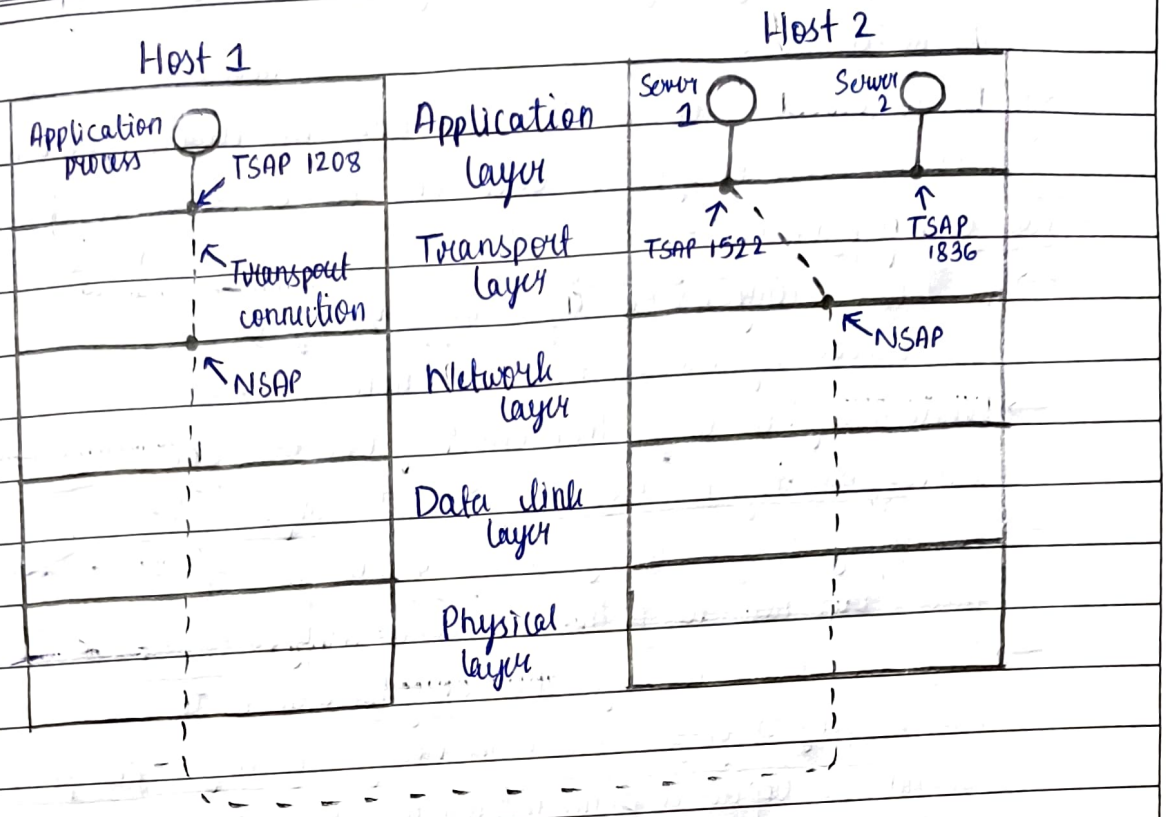
→ The endpoint in the network layer is called 'Network Service Access point' (NSAP).

→ IP Addresses are examples of ~~NAI~~ NSAPs.

Date _____

Page No. _____

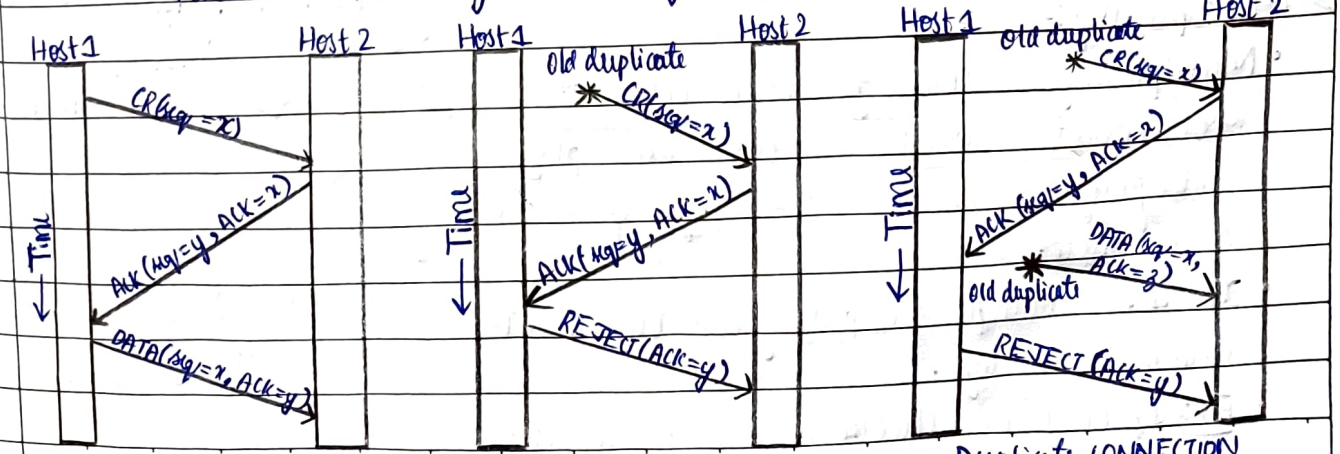
Expt. No. _____



- TSAPs, NSAPs & transport connections -

Connection Establishment :-

- It seems sufficient to just send a CONNECTION REQUEST segment to the destination & wait for a CONNECTION ACCEPTED reply.
- But a problem occurs when network can lose, delay, corrupt/duplicate packets.
- Packet lifetime can be restricted using :-
  → Restricted network design; → Putting a hop counter; → Timestamping each packet.



- Normal operation -

- Old duplicate CONECTION REQUEST appearing out of nowhere -

- Duplicate CONNECTION REQUEST & duplicate ACK -

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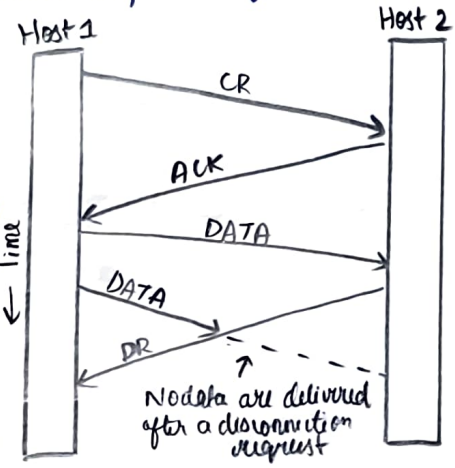
Connection release :-

- There are two styles of terminating a connection :- asymmetric release & symmetric release. (symmetric release is when, one connection is broken, other is broken).
- Symmetric release treats the connection as two separate unidirectional connections and requires each one to be released separately.
- Two army problem :-
  - → The white army is encamped in a valley & surrounded by blue armies on hill sides.
  - → White army is larger than either of the ~~blue~~ blue armies, but together blue army > white army.
  - → So, to defeat white army, the blue armies need to meet, but need to go through white army.
  - → Now, to send a message, their only medium is to send messengers on foot down the valley, but they might be captured & the message may be lost. (i.e. they have to use an unreliable communication channel).



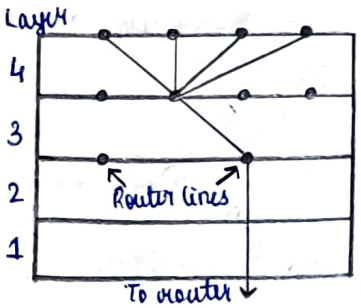
- Abrupt disconnection with loss of data -

Error control & flow control :-

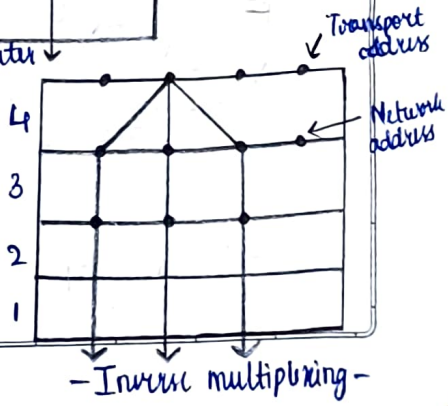
- Key issues are error control & flow control.
- Error control is ensuring that the data is delivered with the desired level of reliability, usually that all of the data is delivered without any errors.
- Flow control is keeping a fast transmitter from overrunning a slow receiver.

Multiplexing :-

- Multiplexing, or sharing several conversations over connections, virtual circuits, and physical links plays a role in several layers of network architecture.
- In TL, the need for multiplexing can arise in a number of ways.
- Ex :- If only one network address is available on a host, all transport connections on that machine have to use it.



- Multiplexing -



- Inverse multiplexing -

Date

Page No.

Expt. No.

INTERNET TRANS. PROTOCOLS :-

UDP :-

- UDP (User Datagram Protocol) is a connectionless transport protocol that sends packets directly blw applications without establishing a connection.
- It provides a simple way to send encapsulated IP datagrams with minimal overhead.
- A UDP segment has an 8-byte header followed by the payload.
- Source & destination ports identify the sending and receiving processes.
- The length field includes header + data
- An optional checksum covers the header, data, and an IP pseudoheader for reliability.

32 bits

Source port	Destination port
UDP length	UDP checksum

- **Remote Procedure Call :-**
  - → Remote Procedure Call (RPC) allows a program to execute procedures on a remote host as if they were local.
  - → Data is exchanged through parameters and return values, hiding message passing from the programmer.
  - → The client stub represents the server procedure on the client side, & the server stub handles execution on the server.
  - → Marshalling is the process of packing parameters for transmission.
  - → Steps in making an RPC :-
    - Step 1: Client calling the client stub. This call is a local procedure call.
    - Step 2: Client stub packing the parameters into a message and making a system call to send the message.

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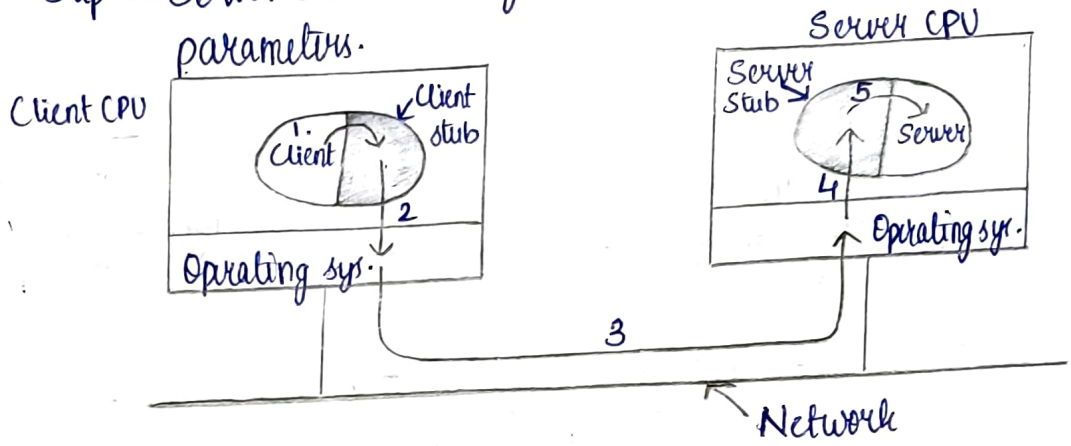
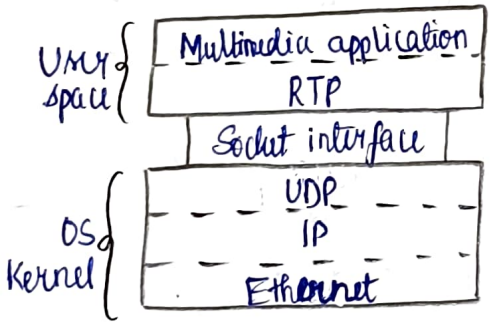
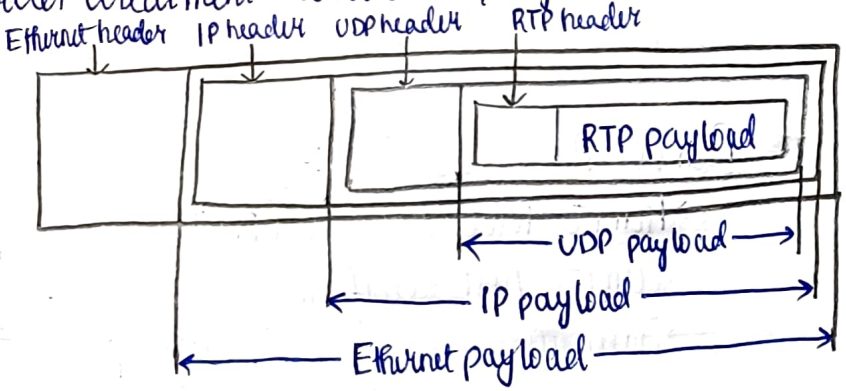
- Step 3: Operating system sending the message from the client machine to the server machine.
- Step 4: Operating system passing the incoming packet to the server stub.
- Step 5: Server stub calling the server procedure with the un-marshaled parameters.



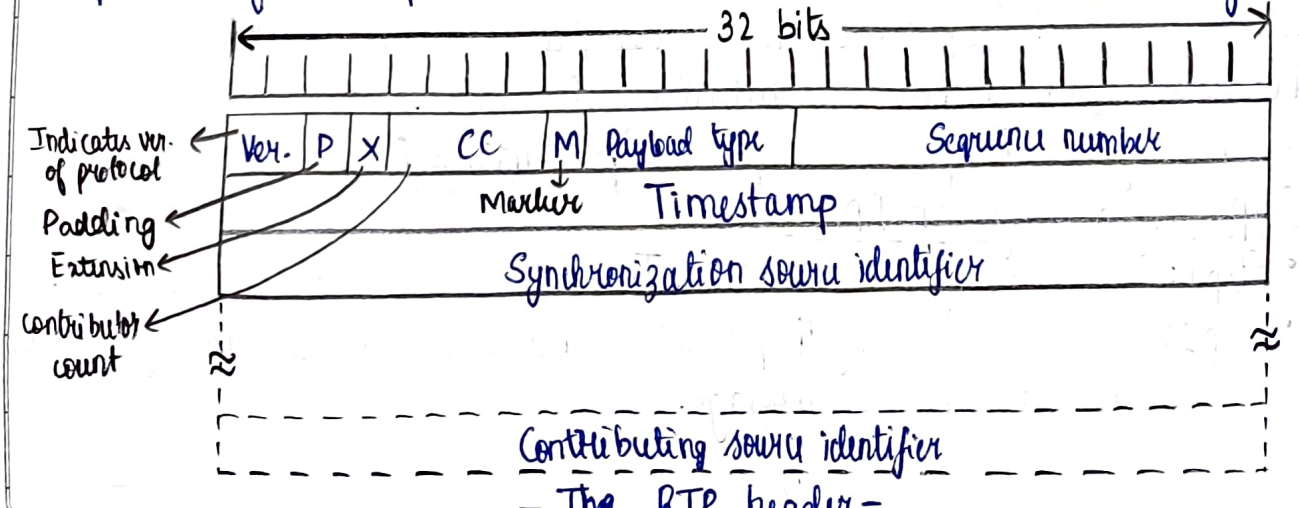
- **Real-time Transport Protocol :-**
  - → RTP (Realtime Transport Protocol) multiplexes multiple real-time data streams into a single UDP stream, which can be unicast / multicast.
  - → RTP packets get no special router treatment unless IP QoS features are enabled.



- position of RTP in protocol stack -



- Packet nesting -



- The RTP header -

Date

Page No.

Expt. No.

→ Version - Indicates the version of the protocol.
→ P(Padding) - Used to indicate if there are extra padding bytes at the end of the RTP packet.
→ X(Extension) - Indicates presence of an extension header b/w the header and payload data.
→ CC(Contributor count) - 4 bit field which indicates no. of contributors.
→ M(Marker) - Signalling used at the application level in a profile-specific manner.
→ PT(Payload type) - Indicates the format of the payload.
→ Seq. number - Seq. number is incremented for each RTP.
→ Timestamp - Used by the receiver to play back the received samples at appropriate time & interval.
→ SSRC - uniquely identifies source of the stream.
→ CSRC - Contributing source IDs enumerate contributing sources to a stream which has been generated from multiple sources.

TCP :-

TCP provides reliable end-to-end communication over unreliable internet works by dynamically adapting to varying network conditions such as topology, bandwidth, delay and packet size, ensuring robustness against failures.

• TCP protocol :-

→ Every byte on a TCP connection has its own 32-bit sequence number.
→ The sending & receiving TCP entities exchange data in the form of segments. A TCP segment consists of a fixed 20-byte header followed by zero/more data bytes.
→ Each link has an MTU (Maximum transfer unit). Modern TCP implementations perform path MTU discovery by using the technique.

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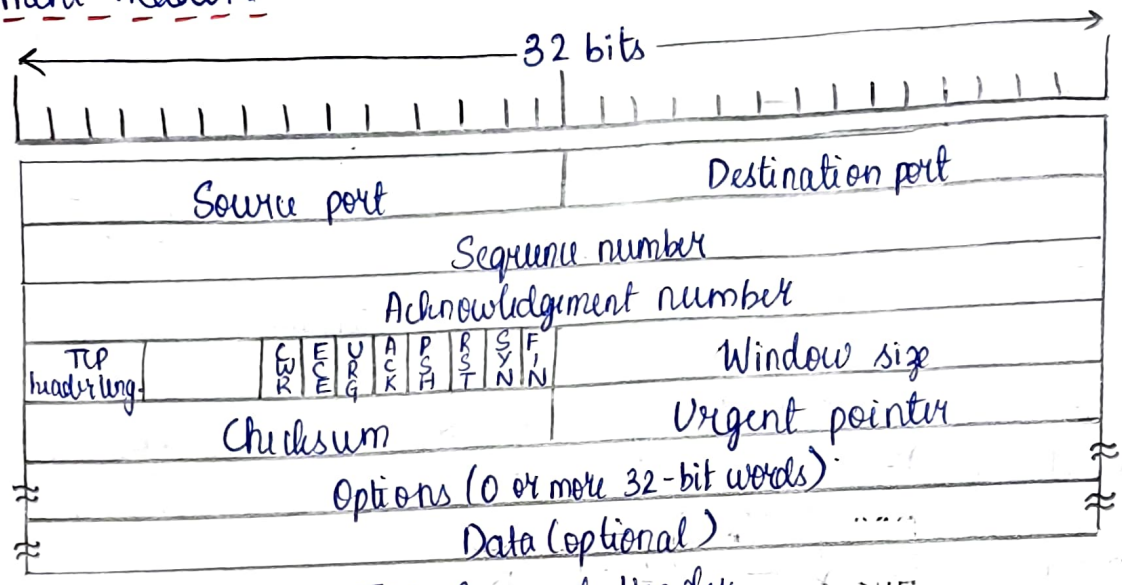
• TCP Segment Header :-

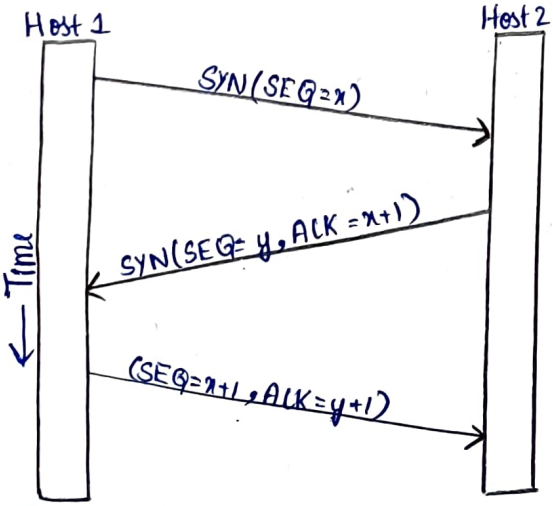


- TCP Segment Header -

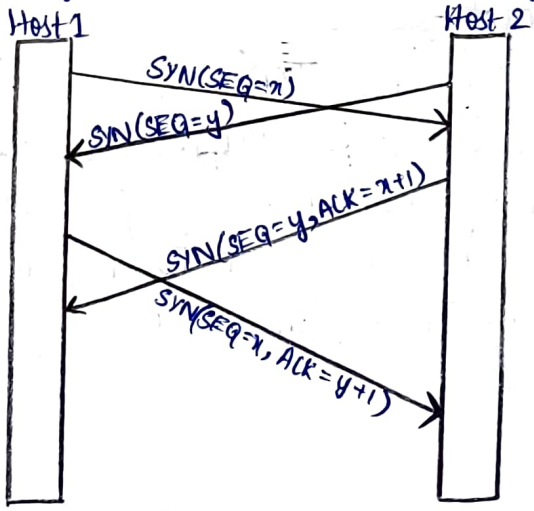
→ CWR (1 bit) - Congestion window reduced (CWR) is set by sending host.
→ ECE (1 bit) - Explicit Congestion Notification.
→ URG (1 bit) - Indicates that the urgent pointer field is significant.
→ ACK (1 bit) - Indicates that acknowledgment field is significant.
→ PSH (1 bit) - Push function.
→ RST (1 bit) - Reset the connection
→ SYN (1 bit) - Synchronize sequence numbers.
→ FIN (1 bit) - Last packet from sender.

• TCP Connection Establishment :-

Connections are established in TCP by means of the three-way handshake.



- TCP connection establishment in normal Case -



- Call collision -

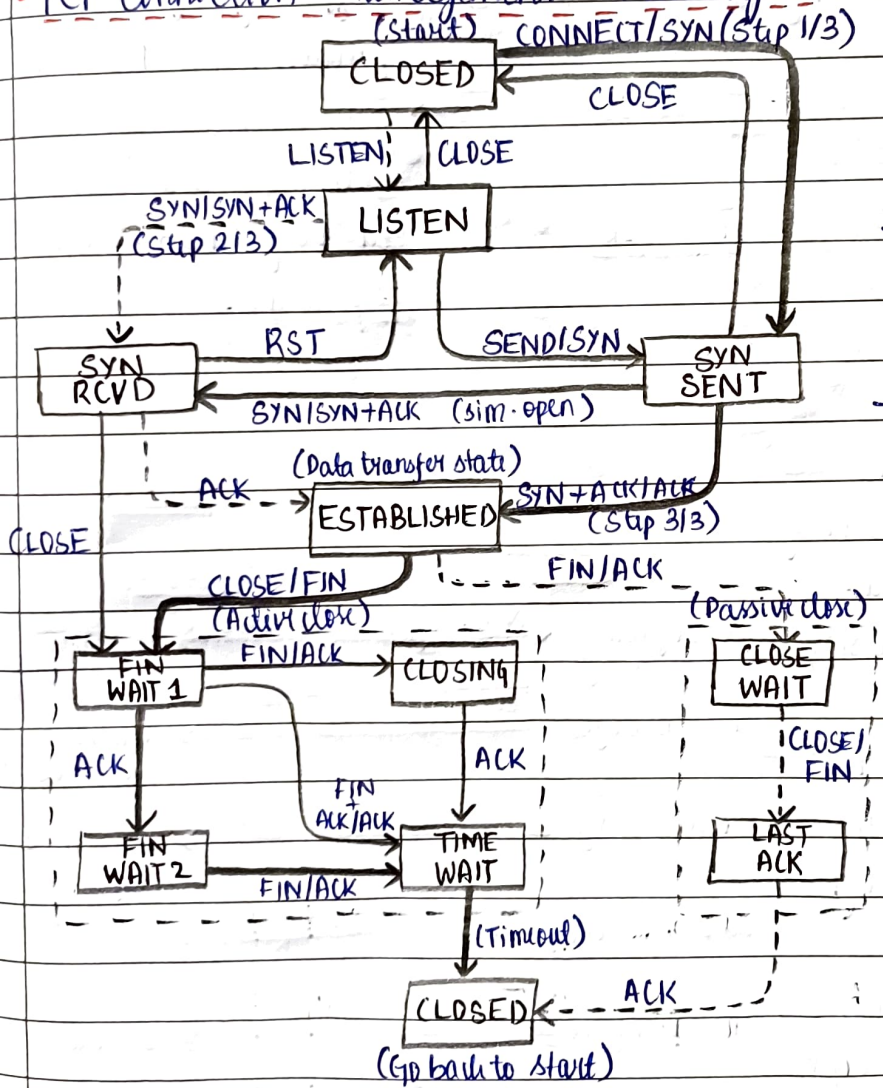
Date _____

Expt. No. _____ Page No. _____

• TCP Connection release :-

TCP connections, though full duplex, are treated as two independent simplex connections. Either side can send a FIN segment to end its transmission; once acknowledged, that direction closes. The connection is fully released only after both sides send & acknowledge FINs.

• TCP Connection Management Modelling :-



→ heavy solid line is the normal path for a client.

→ The heavy dashed lines is normal path for a server.

→ The light lines are unusual events.

→ Each transition is labelled by the event causing it and the action resulting from it, separated by a slash.

PTO →

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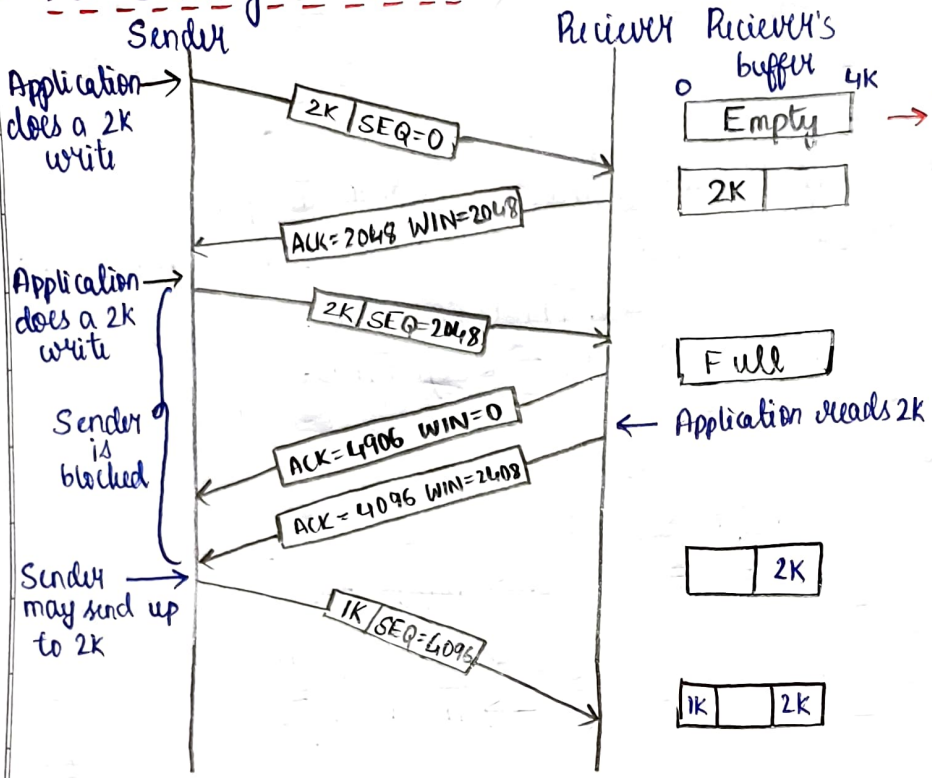
• TCP Sliding Window :-



→ TCP optimizations include delayed acknowledgements, which wait upto 500 ms to combine ACKs and window updates, reducing network load.

→ Nagle's algo: minimizes inefficiency by combining small packets before sending.

→ Silly window syndrome, occurs when data are sent or read in very small chunks, degrading performance.

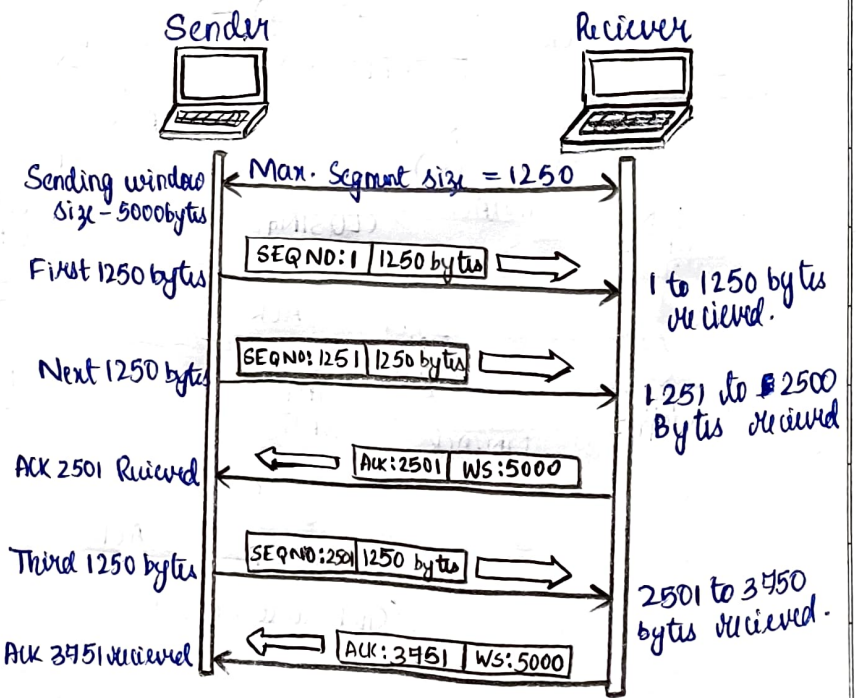
• TCP flow control :-

→ Flow control ensures the reciever process data reliably by adjusting the data flow.

→ It uses the 16-bit window size field in the TCP header.

→ The window size limits how much data the sender can transmit before recieving an acknowledgment.

→ It represents the amount of data the reciever can accept and process at once.



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Expt. No. _____ Page No. _____

• TCP Congestion Control:-

Goal: Prevent network overload by controlling data transmission rate.

→ When congestion builds up: Queues grow large at routers → packet loss → delay.

→ TCP's Role: Uses feedback from the network to slow down sending rate.

→ AIMD (Additive Increase Multiplicative Decrease):

- On acknowledgement, TCP increases congestion window size slowly.
- On packet loss, it reduces the window size sharply.
- Ensures fair and efficient bandwidth use among connections.

→ Mechanism:

- Each ACK → sender increases congestion window by 1.
- On loss → sender halves window.
- Balances efficiency & fairness.

• TCP Timer Management:-

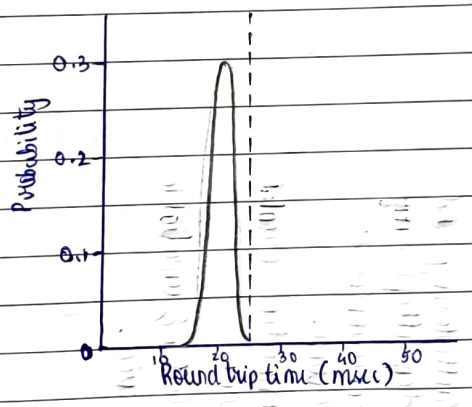
Purpose: Maintain reliability using timers for retransmissions.

→ Main timer: RTO (Retransmission time out)

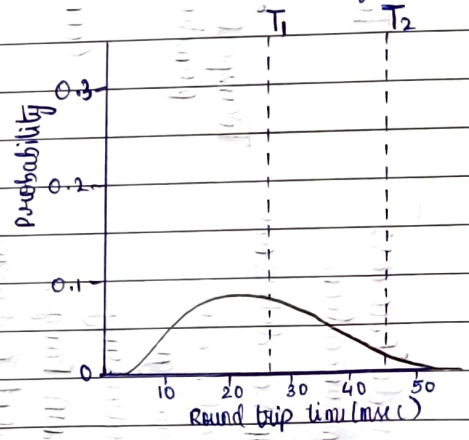
Starts when segment is sent; stops when ACK is received.

→ Challenges:

- Network delays are variable → hard to choose accurate timeout.
- Unlike DLL, TCP's round trip time varies due to routing & congestion.



- Prob. density of ACK arrival times in DLL -



- Probability density of ACK arrival times for TCP -

• Future of TCP:-

→ TCP evolution: Continues adapting for better performance congestion control & robustness.

→ Variations: Many versions exist with improved congestion control algorithms.

→ Challenges:

- TCP cannot meet all application-level transport needs.
- Congestion control still needs optimization for high-speed networks & varying conditions.

→ Outlook: TCP will continue evolving as Int. tech. grows.

MIND MAP - UNIT 4

UNIT - 4

Process to Process Communication

- Types of deliveries
- Client / Server Paradigm
- Addressing
- IANA Ranges
- Socket Addressing
- Multiplexing and demultiplexing
- Connectionless & servic.
- Connection-oriented servic
- Reliable & unreliable servic.

Transport Services

- Transport servic primitives
- Berkeley sockets

Elements of Transport Protocols

- Multiplexing
- Addressing
- Connection establishment
- Error control & flow control
- Connection release

Internet Transport Protocols

- UDP
  - Remote procedure call
  - Realtime transport protocol
- TCP
  - TCP sliding window
  - TCP protocol
  - TCP Segment holder
  - TCP conn. establishment
  - TCP connection release
  - TCP flow control
  - TCP conges. control
  - Future of TCP
  - TCP Connection Management modelling

Date _____

Page No. _____

Expt. No. _____

UNIT-5

Syllabus:-

Application layer: DNS - The Domain Name System, The DNS Name Space, Domain Resource records, Name Servers.

Elec. MAIL: Architecture & Services, The User Agent, Message Formats, Message transfer, Final delivery.

The World Wide Web: Architectural Overview.

Streaming Audio & Video: Streaming stored media, Streaming live media, Content delivery.

DOMAIN NAME SYS.:

- The DNS resolves high-level human readable names for computers to low-level IP addresses.
- The essence of DNS is the invention of a hierarchial, domain-based naming scheme and a distributed database system for implementing this naming scheme.
- To map a name onto IP address, an application program calls a lib. procedure called the resolver, passing it the name as a parameter.
- The resolver sends a query containing the name to a local DNS server.
- The query and response messages are sent as UDP packets.

➢ The DNS Name Space:-

→ The top of the naming hierarchy is managed by an org. called ICANN. [Internet Corporation for Assigned Names and Numbers].

→ The Internet is divided into over 250 top-level domains.

→ All these domains can be represented by a tree - where a leaf domain may contain a single host.

→ The top level domains come in two types:- generic & countries.

.com, .edu, .ac etc. ← generic

countries → .in, .us, .au

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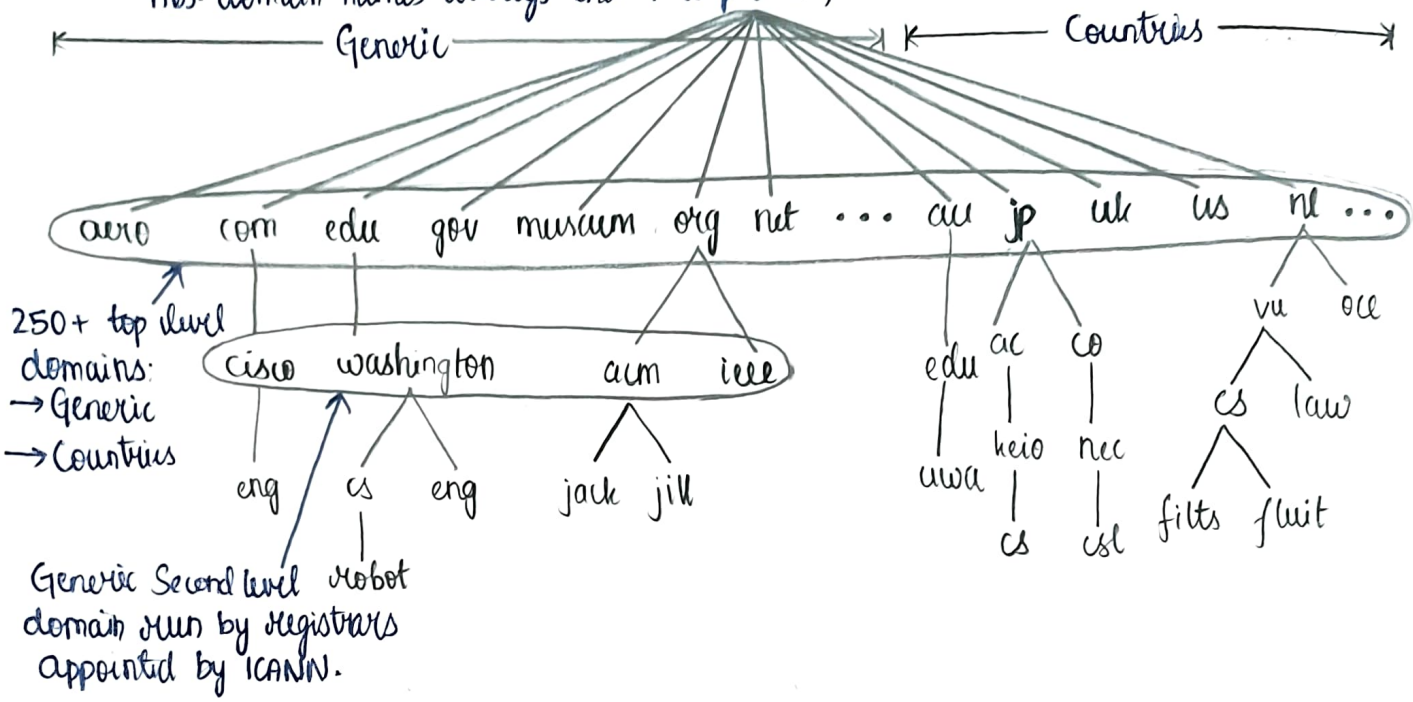
→ DNS Namespace is hierarchial from the root down.
- different parts delegated to diff. organisations.

- Can register under multiple top-level domains.
- Abs. domain names always end in a period, relative domain names do not.



Domain Resource records:-

- Every domain (host or top-level) has resource records - the DNS database.
- For a host, the main record is its IP address, though other types exist.
- DNS returns the resource records for a given domain name.
- DNS mainly maps domain names to resource records.
- A resource record is a five-tuple, usually shown as one line of ASCII ~~text~~ text.
  → Domain_name, TTL, Class, Type, Value.
- The key resource records in the namespace are IP addresses and name servers.

Name Servers:-

- In theory, a single name server could contain the entire DNS database and respond to all queries about it.
- To avoid the problems associated with having only a single source of information, the DNS name space is divided into non-overlapping zones.
- Name servers contain data for portions of the name space called 'zones'.
- Finding the IP address for a given hostname is called 'resolution' and is done with the DNS protocol.

★ DNS protocol - Runs on UDP port 53, retransmits lost messages. Caches name server answers for better performance.

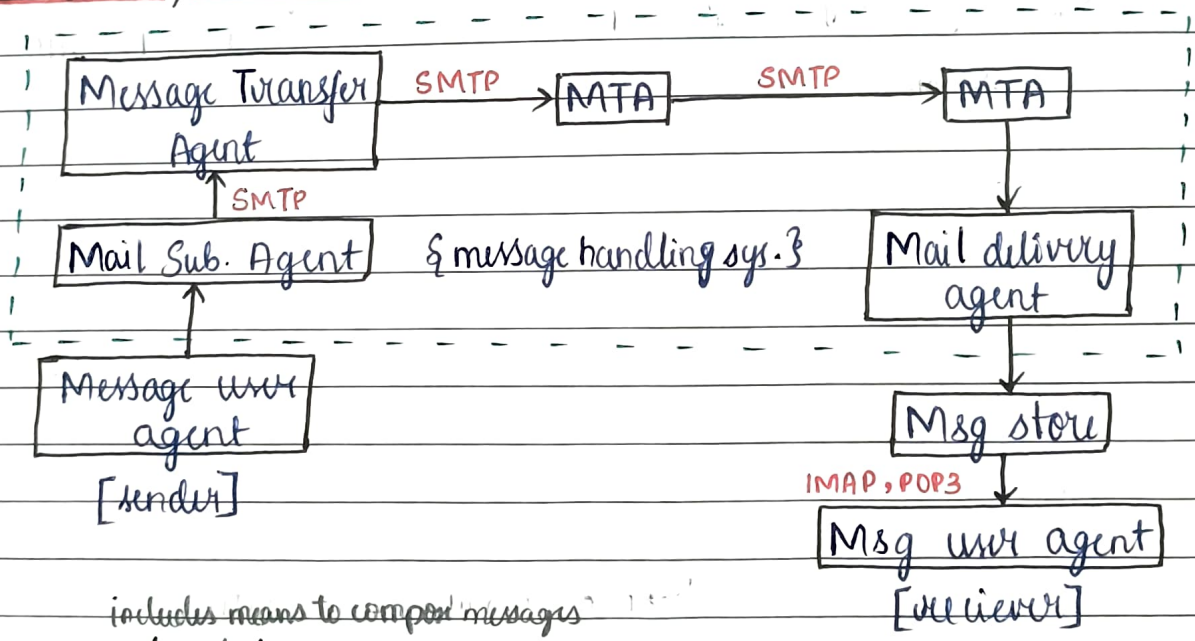
Expt. No. _____ Page No. _____

Computer requests local name server to resolve → Local name server asks root name server → Root returns the name server for lower zone → Continue down zones until a name server can answer.

ELECTRONIC MAIL :-

- Most widely used ~~serve~~ servies of Internet.
- Messages in mail not only contains text, but also images, audio & video data.
- Sending mail - sender ; Recieving mail - reciever.

Architecture & Services :-



includes means to compose messages and reply to messages.

- The user agent is a program that provides a graphical interface, that lets user interact with email system.
- The message transfer agents are typically system processes. Their job is to automatically move email through the system from originator to reciepent with SMTP.

The User Agent :-

What users see - interface elements of a typical user agent.

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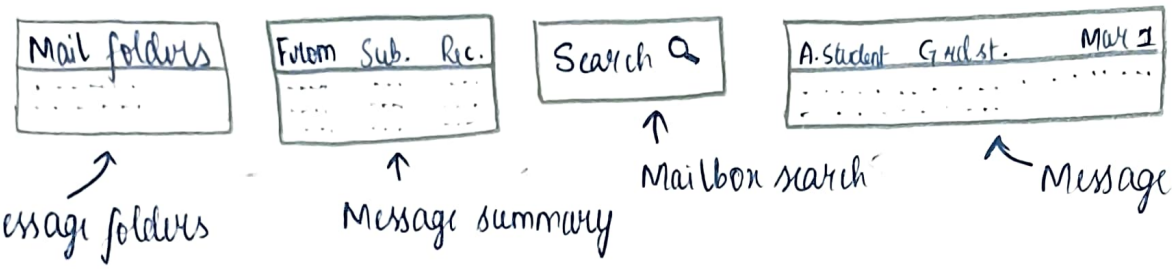


Message formats :-

Header	Meaning
To:	Email address of primary recipient
Cc:	Email address of secondary recipient
Bcc:	Email add. of blind carbon copy.
From:	Person who created the message
Sender:	Email address of actual sender
Received:	Line added by each TA along the route
Return-Path:	Can be used to identify a path back to sender.

- Header fields related to message transport -

Header	Meaning
Date:	Date & Time message was sent.
Reply-to:	Email id to which replies should be sent.
Message-Id:	Unique no. for referencing this message.
In-Reply-To:	Mess. Id to which this is a reply.
References:	Other relevant message Ids.
Keywords:	User-chosen keywords.
Subject:	Short one-line summary of message.

- Other header fields -

Header	Meaning
MIME-Version:	Identifies the MIME version.
Content-description:	Human-readable string telling what is in the message.
Content-Id:	Unique identifier
Content-Transfer-Encoding:	How the body is wrapped for transmission.
Content-type	Type & format of the content.

- MIME header fields used to describe what content is in body of the message -

- Common MIME content types and subtypes -

Type	Example subtypes	Description
text	plain, html, xml, css	Text in various formats
image	gif, jpeg, tiff	Pictures
audio	basic, mpeg, mp3	Sounds
video	mpeg, mp4, quicktime	Movies
model	vrml	3D model
application	octet-stream, pdf, javascript, zip	Data produced by applications
message	http, rfc822	Encapsulated message.
multipart	mixed, alternative, parallel, digest	Combination of multiple types.

Date

Page No.

Expt. No.

Message Transfer :-

Messages are transferred with SMTP (Simple Mail transfer protocol).

- Readable text commands
- Submission from user agent to MTA on port 587.
- One MTA to the next MTA on port 25.
- Other protocols for final delivery (IMAP, POP3)

Final delivery :-

User agents ~~uses~~ uses protocol like IMAP for final delivery.

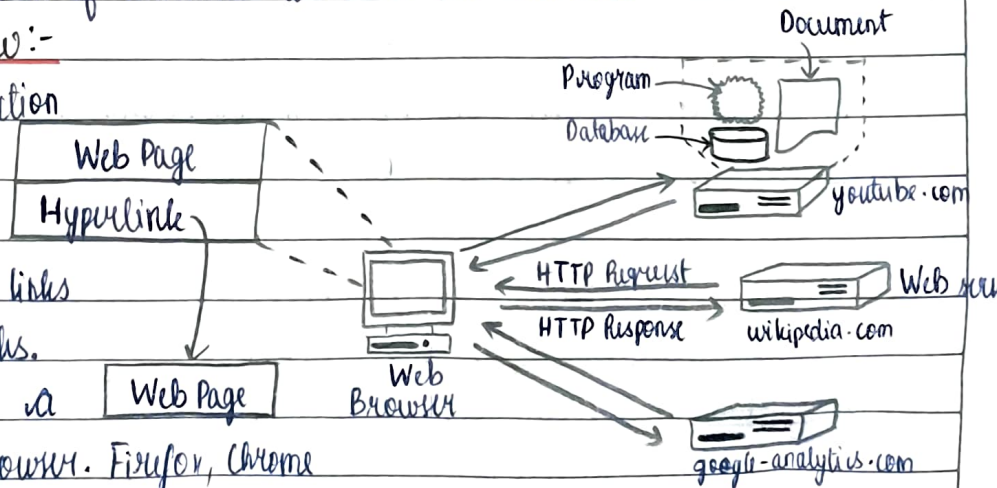
- Has commands to manipulate folders / messages.
  Ex:- EXAMINE, CREATE, DELETE, RENAME, SUBSCRIBE, LIST, LSUB etc.
- Alternatively, a Web interface (with proprietary protocol) might be used.

WORLD WIDE WEB :-

- The World Wide Web, is an architectural framework for accessing linked content.
- Spread out over millions of machines all over the internet.

Architectural Overview :-

- Web - worldwide collection of content in the form of ~~tot~~ web pages.
- Each page may contain links to other pages - hyperlinks.
- Pages are viewed with a program called - browser. Firefox, Chrome etc. are popular browsers.
- Pages are named with URLs (Uniform Resource Locators).



http://www.phdcomics.com/comics.php

Protocol — Server — Page on server

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- Steps a client takes to follow a hyperlink:-
  - Determine the protocol (HTTP)
  - Ask DNS for IP address of server.
  - Make a TCP connection to server.
  - Send request for the page; server sends it back.
  - Fetch other URLs as needed to display the page.
  - Close idle TCP connections.
- Steps a server takes to serve pages:-
  - Accept a TCP connection from client.
  - Get page request and map it to a resource (eg. file name).
  - Get the resource (eg. file from disk)
  - Send contents of the resource to the client.
  - Release idle TCP connections.

HTTP:-

HTTP (HyperText transfer Protocol) is a request-response protocol that runs on top of TCP.

- Fetches pages from server to client.
- Server usually runs on port 80.
- Headers are given in readable ASCII
- Content is described with MIME types.
- Protocol has support for pipelining requests.
- Protocol has support for caching.

STREAMING AUDIO & VIDEO:-

two or more media, to be played during some well-defined time-interval

Streaming audio & video refers to continuous transmission of multimedia data over the internet in real time.

→ Unlike file downloads, users don't wait for the entire file to arrive - playback begins as soon as enough data is buffered.

→ Why streaming?- Users prefer instant playback; supports live events, interactive communication etc

→ Applications:-
- Stored streaming - Youtube, Netflix, Prime Video.

Expt. No. _____ Date _____ Page No. _____

- Live streaming - Sports, online radio, Twitch etc.
- Interactive streaming - Zoom, Gmeet, MS Teams.

➤ Digital Audio :-

- Sound is a pressure wave in air.
- Digital audio represents this wave in binary form.
- An ADC (Analog-to-Digital Converter) converts analog voltage to binary values, while a DAC (Digital-to-Analog Converter) does the reverse, producing analog voltage.
- A loudspeaker then converts this voltage into sound.
- The error caused by limited bits per sample is called quantization noise - if large, it becomes audible.
- Sampling & Quantization :-
  Sampling - Measure the sound wave at discrete intervals.
  Quantization - Represent each sample by a finite number of bits.

★ Compression of Audio :-

Raw audio files are large, so compression is essential to reduce size, bandwidth, and transfer time.

→ Compression uses two algorithms: encoding (at source) & decoding (at destination).

→ Popular compressed formats include MP3 and AAC, used in MPEG-4.

→ Lossless Compression :-
- No data loss
- Formats: FLAC, ALAC, WAV
- Used where quality is crucial (eg. archiving, editing).

→ Lossy Compression :-
- Removes inaudible frequencies
- Much smaller file sizes.
- Formats: MP3, AAC, OGG, Opus.
- MP3 uses perceptual coding.

Teacher's Signature _____

Digital Video:-

- Raw video is huge, so compression is critical.
- Video = sequence of still images shown rapidly, to create motion.

★ Compression of Video:-

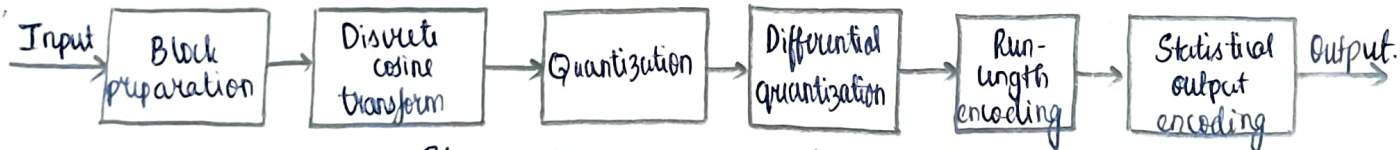
→ Spatial Compression:
- Reduces redundancy within a frame (like JPEG)
- Removes similar color patterns.

→ Temporal Compression:
- Reduces redundancy across frames.
- Stores only changes (motion estimation).

The JPEG Standard:-

JPEG is a standard for compressing still images developed by the Joint Photographic Experts Group. JPEG uses lossy compression - meaning that some image data is discarded to achieve significant size reduction.

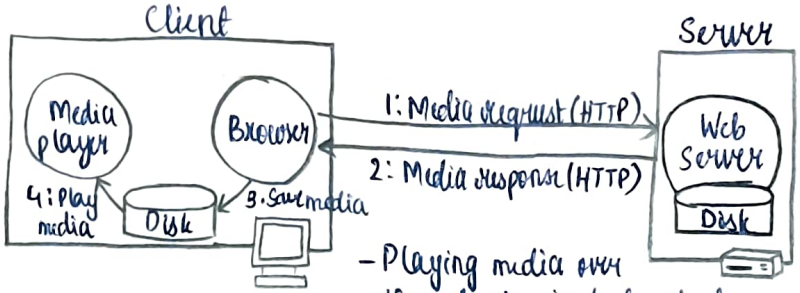
- The standard aims to:-
  - Achieve high compression ratios.
  - Maintain acceptable visual quality.
  - Provide device and platform-independent image representation.
- JPEG exploits two main chars. of human vision:
  - The eye is less sensitive to small color changes than to brightness variations.
  - The eye cannot perceive fine detail in high frequencies.



- Steps in JPEG sequential lossy encoding -

Streaming Stored media:-

- Traditional download model:-
  - Entire file is downloaded before playback.
  - Long wait times for large files - poor user experience.

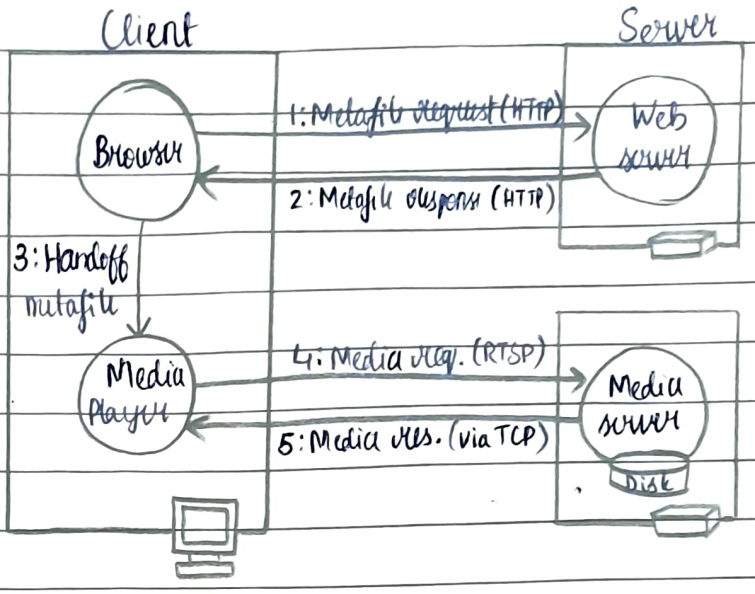


- Playing media over the web via simple downloads -

Date _____

Expt. No. _____ Page No. _____

- Streaming model:
  - Starts playback while downloading.
  - Data is divided into small chunks.
  - Client buffers a few seconds before starting playback.
  - Working:-
    - User clicks a video link.
    - A metafile - contains info like - server add., port, code type etc.
    - Media player reads the metafile & requests the actual stream.



- Streaming media using web & server -

Streaming live media :-

Transmits real time events as they happen; eg - live concerts, cricket matches etc.

- Uses protocols like RTP, RTCP & RTSP for synchronised & controlled delivery.
- Key characteristics:-
  - No stored file - data is captured, encoded, transmitted instantly.
  - Uses low latency transmission protocols.
  - May still use buffering to handle jitter.
- Live streaming architecture:-
  - Capture - Camera/microphone captures raw data.
  - Encode - Compressed using codecs.
  - Send - Streamed to a media server
  - Distribute - Server sends unicast/multicast to viewers.
  - Playback - Player decodes and plays stream.

Real time Conferencing :-

Interactive communication where audio/video is exchanged both ways in real time.

Teacher's Signature _____

Date _____

Expt. No. _____ Page No. _____

- **Protocols :-**
  - SIP : Setup, modify, terminate sessions.
  - RTP/RTCP : Carry & monitor media streams.
  - STUN/TURN/ICE : NAT Traversal and peer connection.
  - WebRTC : Browser-based real-time communication framework.
- **Buffering Vs Latency tradeoff :-**

More buffering	Less Buffering
Smooth playback	Risk of jitter
Higher delay	Lower delay
Good for video streaming	Needed for real-time conferencing.

Category	Examples	Protocols	Challenges	Solutions
Stored streaming	Ytube, Netflix	RTSP, DASH, HTTP	Buffering, delay	Adaptive bitrate
Live streaming	IPTV, Twitch	RTP, RTMP	Jitter, packet loss	Buffer + FES
Real time conferencing	Zoom, Skype.	RTP, SIP, WebRTC	Latency, sync	QoS, small buffers

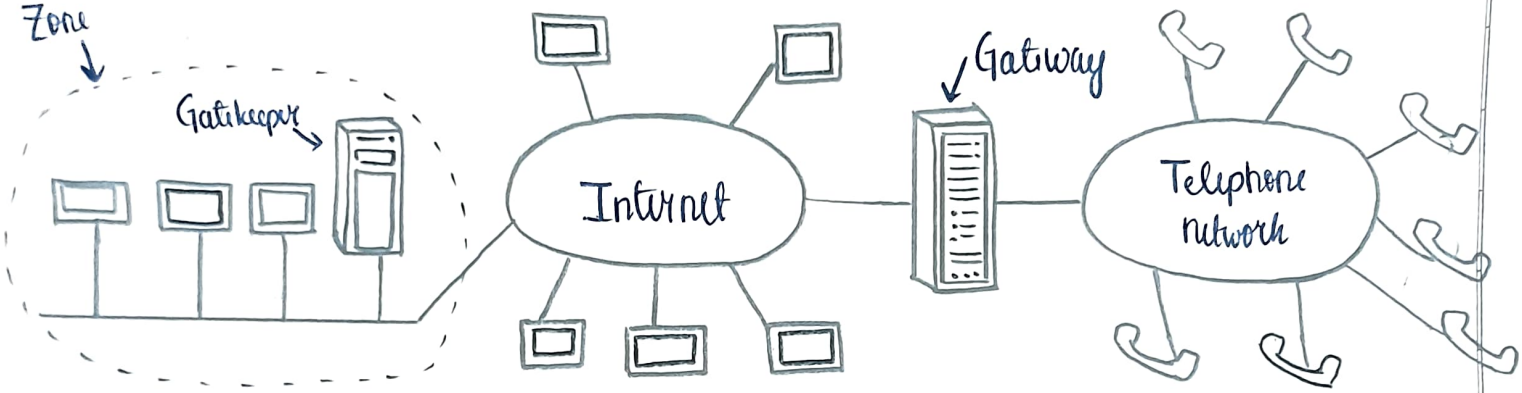
★ H.323 Protocol :-

H.323 is a set of standards for real-time communication over IP networks, such as the internet.

→ Developed by ITU as a way to enable voice, video and data communication over IP networks.

→ H.323 is widely used in a variety of applications, including voice over (VoIP), video conferencing and streaming media.

→ Often used in conjunction with other protocols, such as Session Initiation Protocol (SIP), to deliver audio and video content over the internet.



- The H.323 arch. model for internet telephony -

→ At the centre, is a gateway that connects the Internet to the telephone network.
→ The communicating devices are called terminals.
→ A LAN may have a gatekeeper, which controls the end points under its jurisdiction, called a zone.

MIND MAP - UNIT 5

UNIT-5

- Domain Name System
  - The DNS Name Space.
  - Domain resource records.
  - Name Servers.
- WorldWide Web
  - Architectural Overview
    - Steps to follow a hyperlink.
    - Steps a server takes to serve pages.
  - HTTP
- Electronic mail
  - Architecture & services
  - User Agent
  - Message Format
    - Header fields I
    - Header fields II
    - MIME Header fields
    - Common MIME content types & subtypes.
  - Message Transfer
  - Final delivery.
- Streaming Audio & Video
  - Applications
  - Real time Conf.
    - Protocols
    - H.323
    - Buffering Vs Latency tradeoff.
  - Digital Audio
    - Sampling & Quantization
    - Compression of Audio
      - Lossless Comp.
      - Lossy compression
  - Digital Video
    - Compression of Video
      - Spatial
      - Temporal
  - The JPEG Std.
  - Streaming stored media
    - Traditional download model
    - Streaming model
  - Streaming live media
    - Key characteristics
    - Live streaming Architecture